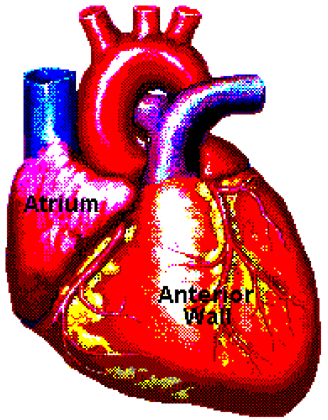
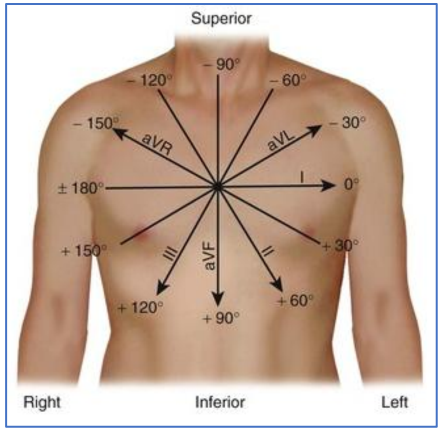


Basic Principles of Medicine 1

Module: Cardiovascular, Pulmonary, Renal

Lecture: CPR 13



Origin and Components of Electrocardiogram (ECG)

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Recommended Readings

Chapter 12: Electrical Activity of the Heart: The Electrocardiogram. Medical Physiology: Principles for Clinical Medicine, 6e. Rodney A. Rhoades, David R. Bell.

Chapter 11: Fundamentals of Electrocardiography: Guyton and Hall Textbook of Medical Physiology, 14th Edition. John E. Hall PhD and Michael E. Hall MD, MS.

Slides with blue background are the supplemental information of the previous slides and thus, might not be explained again during the lecture time. However, please read the slides for clarification as they are equally important to know

Learning Objectives

SOM.MK.II.BPM1.3.CPR.2.PHYS.0171 Explain the concept of dipole, vector of depolarization/repolarization, and how these vectors generated in the heart produce the waveforms of the ECG.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0172 Name the parts of a typical bipolar (Lead II) ECG tracing and explain the relationship between each of the waves, intervals, and segments in relation to the electrical state of the heart.

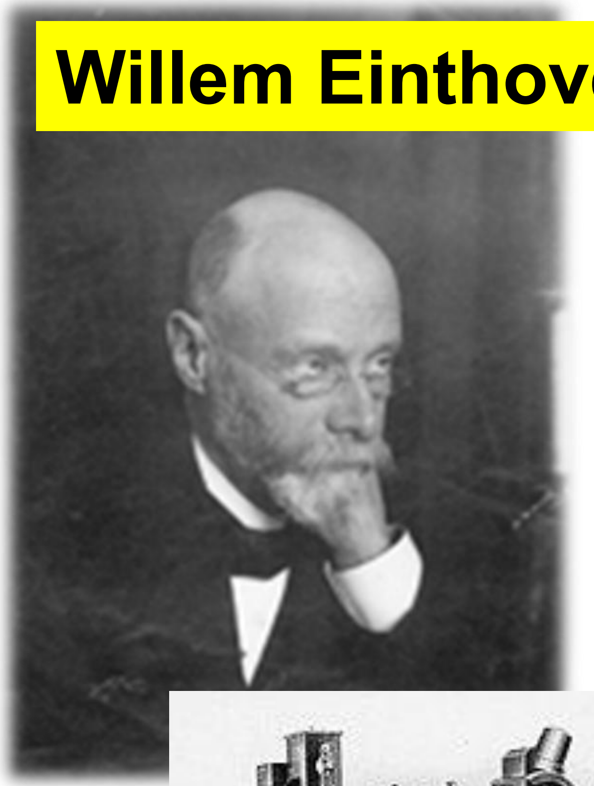
SOM.MK.II.BPM1.3.CPR.2.PHYS.0173 Relate the ECG trace to the different parts of the cardiac action potential.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0174 Explain the electrocardiographic changes associated with the myocardial ischemia, injury and cell death; and the concept of injury current and its relationship with the ST segment elevation or depression.

Lecture Outline

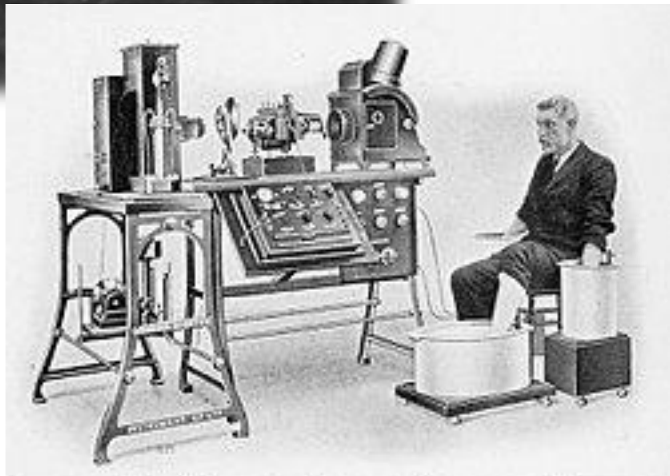
- ❖ Introduction to ECG, Leads & Paper
- ❖ Concept of dipole, wave of depolarization & vector
- ❖ Generation of ECG waves in Lead II
- ❖ Normal ECG waves, intervals & segments
- ❖ ECG related to Cardiac Action Potential
- ❖ How to read an ECG
- ❖ ECG changes in Myocardial ischemia/Infarction

Willem Einthoven (1860 – 1927)



Dutch Physician and Physiologist

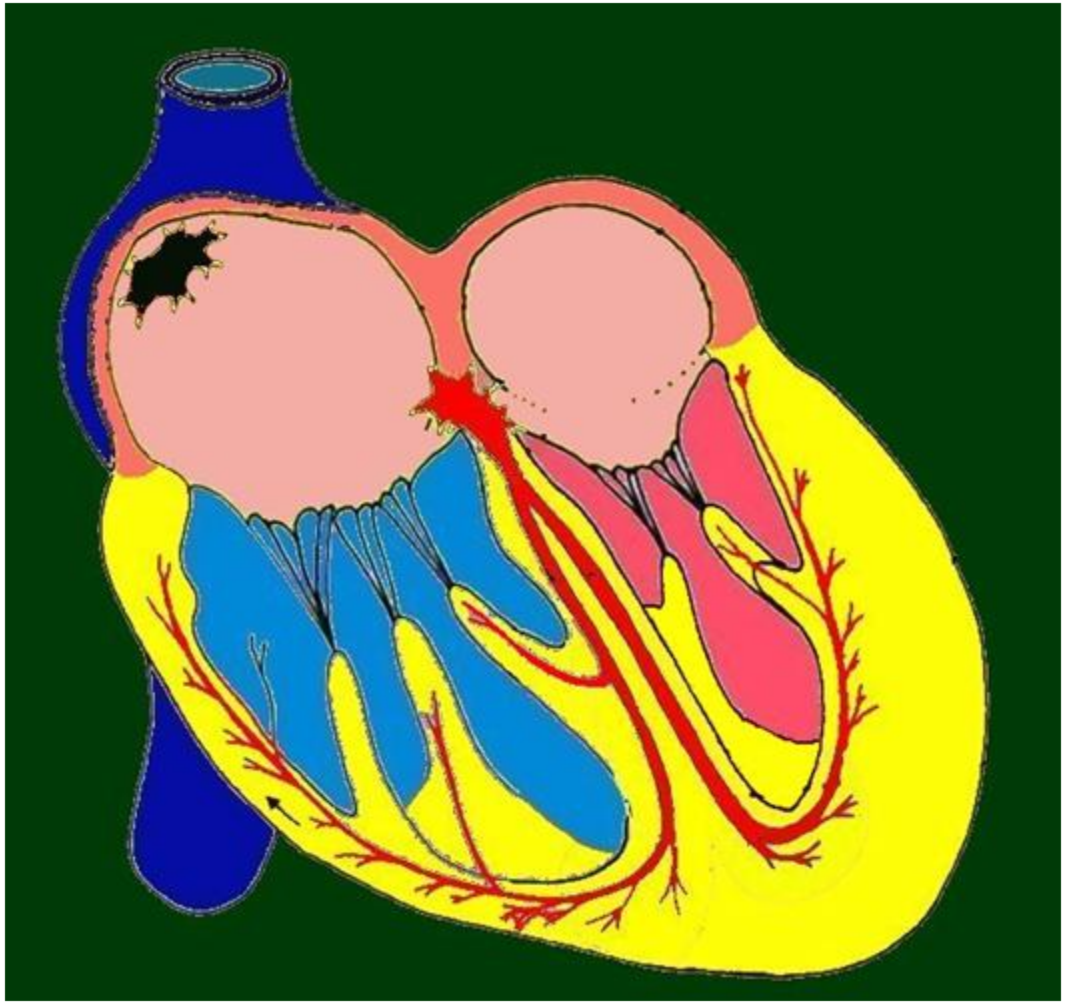
Invented the First practical electrocardiogram machine that recorded ELECTROCARDIOGRAM (ECG/EKG)



STRING GALVANOMETER MODEL

DIGITAL ECG MACHINE

Sequence of spread of action potentials in the Heart



AP generation in the SA node

Right/left atrium depolarization
➔ Atrial contraction

AV nodal transmission (with a delay)

Bundle of His and bundle branches

Septum depolarization

Ventricle walls depolarization
➔ Ventricular contraction

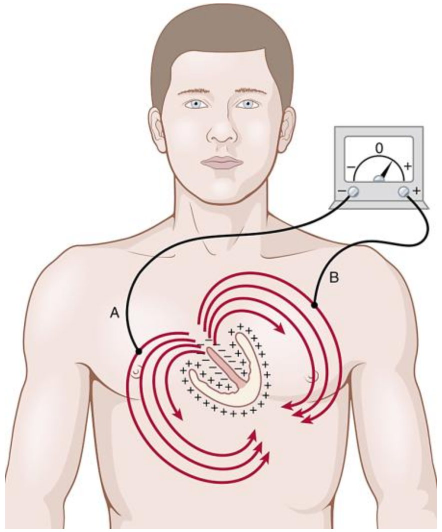
Atria repolarization (*atria relaxation*) occurs during ventricular depolarization

Electrocardiogram (ECG): Definition

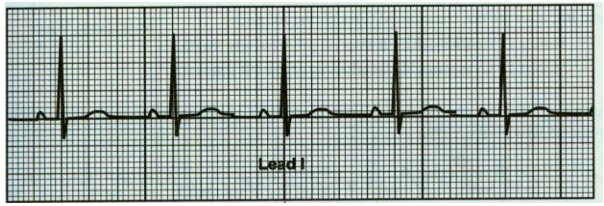
An ELECTROCARDIOGRAM (ECG) is an amplified, timed recording of the SUMMATED electrical activity of the heart generated during every cardiac cycle, as detected on the surface of the body.

Body serves as a volume conductor

The heart generates a collection of changing charges (dipoles) during depolarization and repolarization of the myocytes



Lead is an electrode placed at a ***specific site*** on the body surface to record ECG



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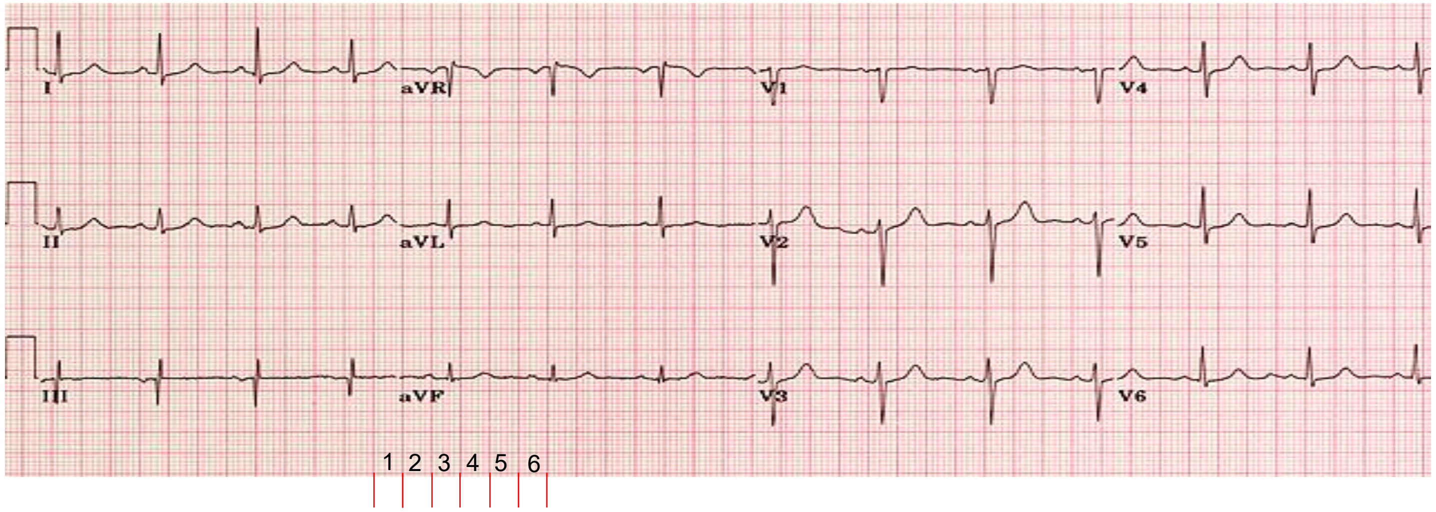
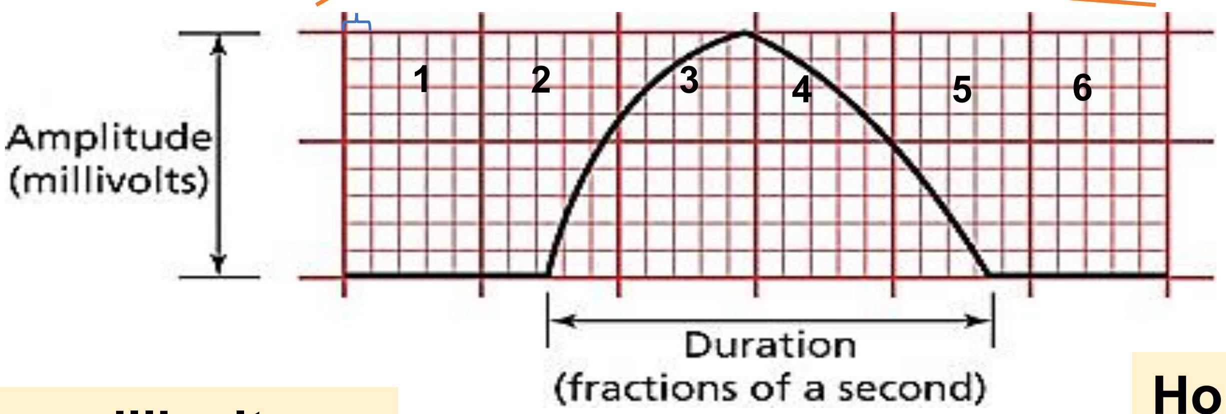
ECG is recorded on a Calibrated graph paper



Recording speed: 25 mm/sec

1 mm = 0.04 sec



Darker lines - Big squares
Lighter lines - Small squares

Vertical axis: millivolts
- 1 small square = 0.1 mV

Horizontal axis: Time
- 1 small square = 0.04 seconds
- 1 big square = 5 small squares
- 1 big square = 0.20 sec

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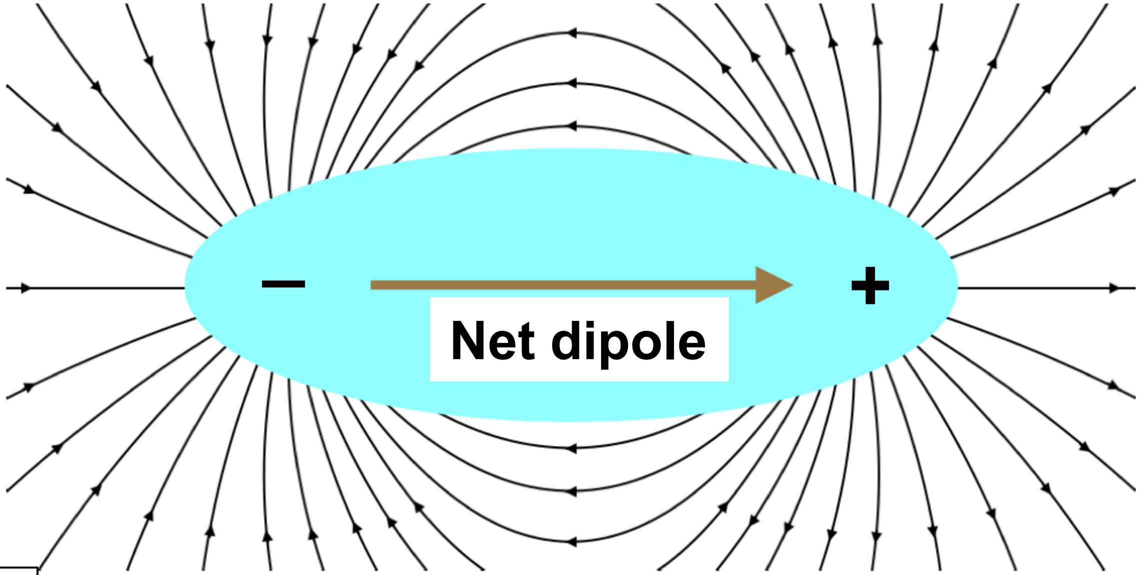
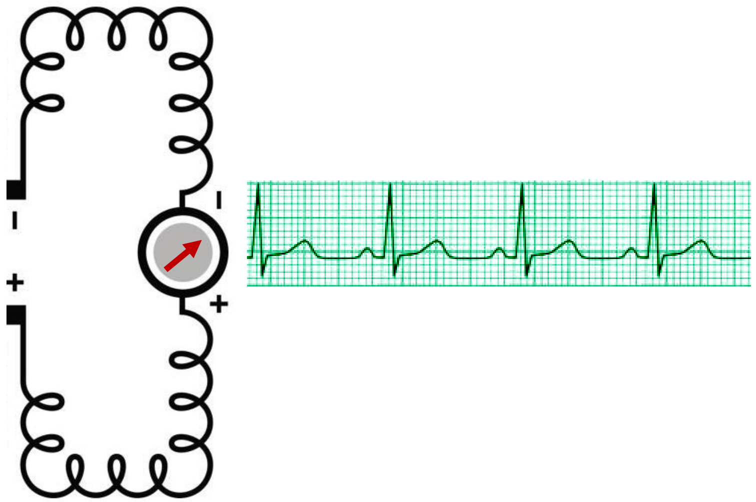
ORIGIN OF ECG WAVEFORMS

During DEPOLARIZATION Or REPOLARIZATION of the myocardium → DIPOLES originate → Electrodes (LEADS) record as ECG





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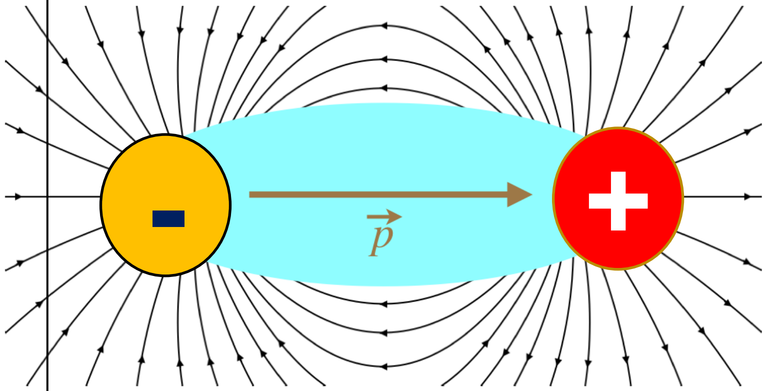
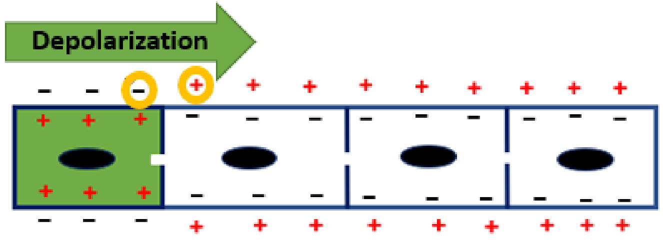
When is a Dipole formed?

The difference of polarity between two neighboring locations is called Dipole.
When a portion of myocardium becomes depolarized from an Action Potential, its polarity is temporarily reversed, becoming positive on the inside and negative on the outside relative to the neighboring regions of opposite charge, or polarity, within the myocardium.



When is a dipole formed?

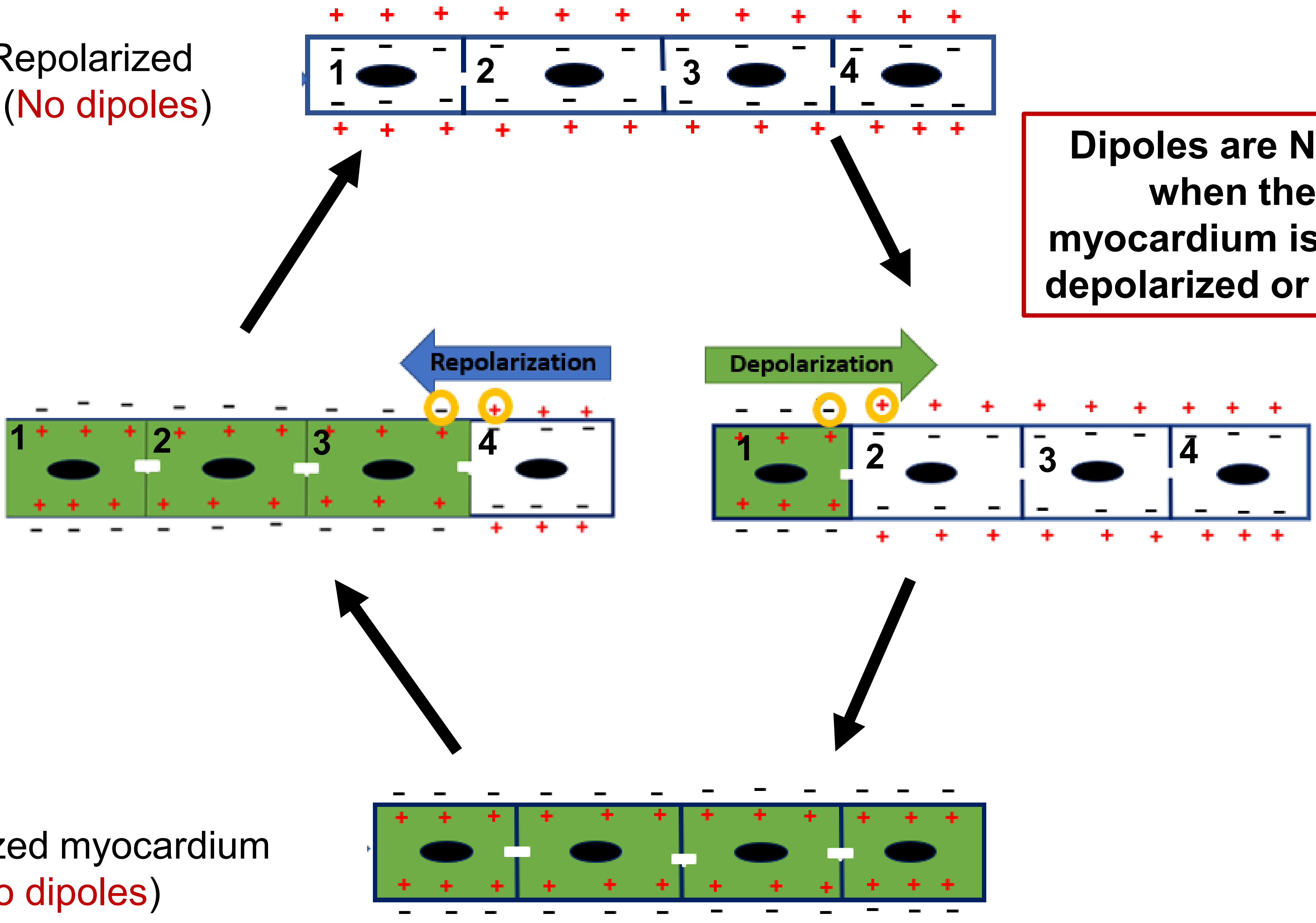
- Dipoles are present only when a portion of the myocardium is in the **process of** depolarization or repolarization while other portions are not.



Resting or Repolarized myocardium (No dipoles)

Dipoles are NOT formed when the entire myocardium is completely depolarized or repolarized.

Dipoles present here

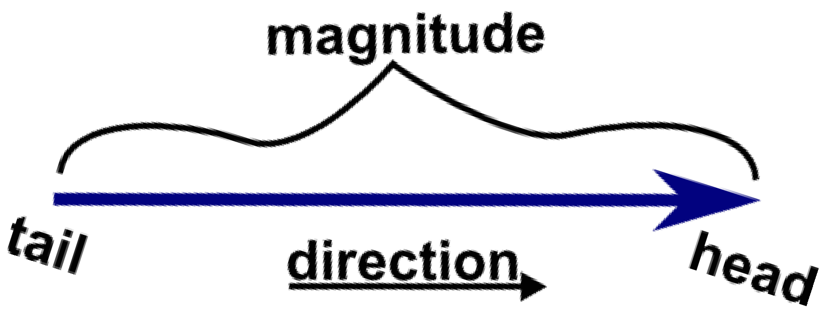


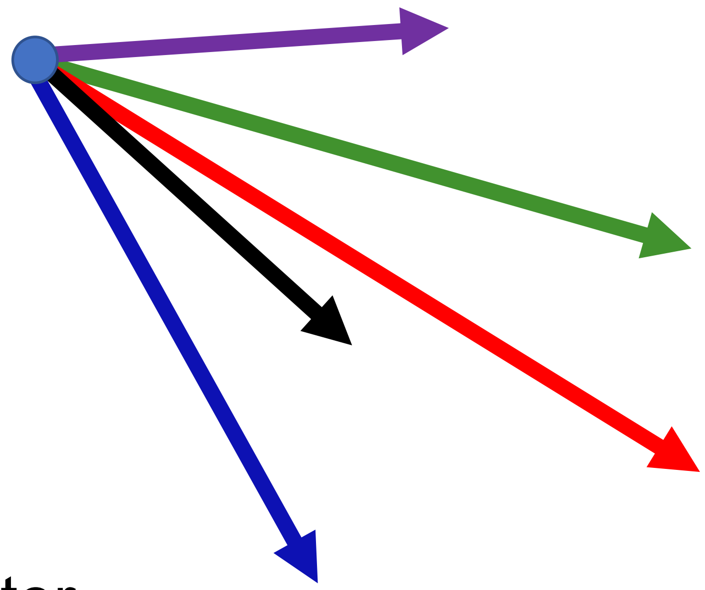
Dipoles present here

Depolarized myocardium (No dipoles)

Dipoles and Vectors

The graphical representation of the magnitude and direction of a dipole is called Electrical VECTOR. A vector has both magnitude and direction.





Several electrical vectors may be represented by ONE resultant vector called **MEAN ELECTRICAL VECTOR** that is recorded by the ECG Leads

RESULTANT Vector (the sum of many vectors)

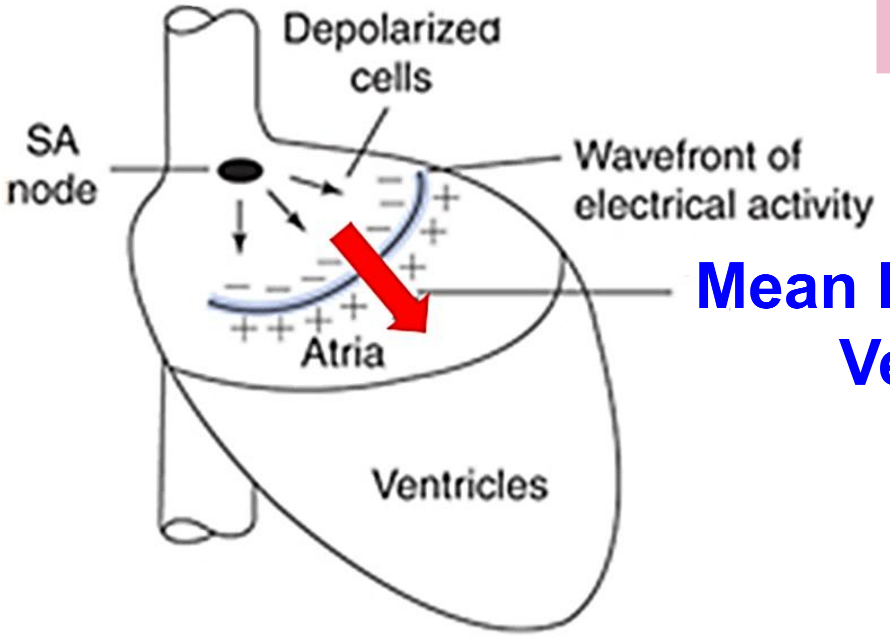
Leads record the electrical activity of the HEART

Lead:
Electrode connection that records the potential difference between 2 points

Negative electrode

Positive electrode

Lead Axis



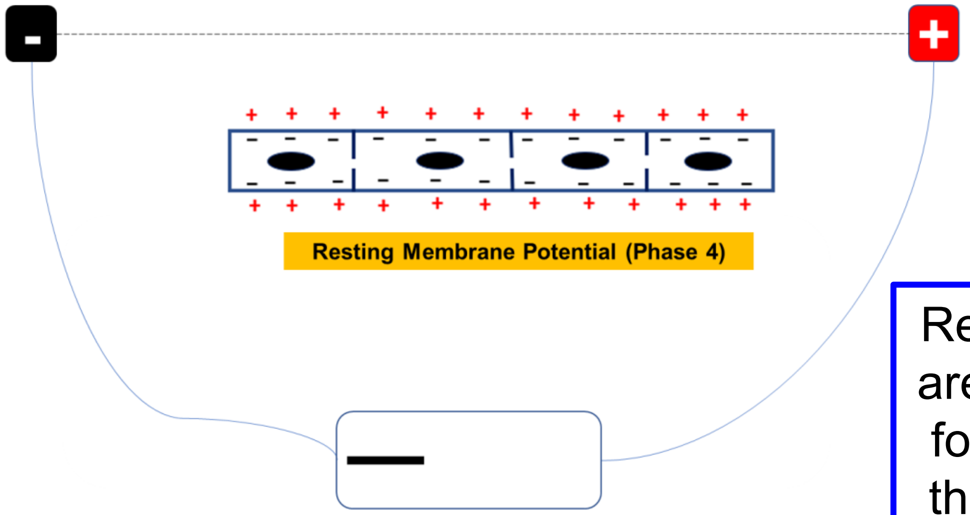
Mean Electrical Vector

Positive Electrode records waves in ECG

Mean Electrical Vector of the ventricular depolarization is also known as Mean Electrical Axis (MEA) of the heart

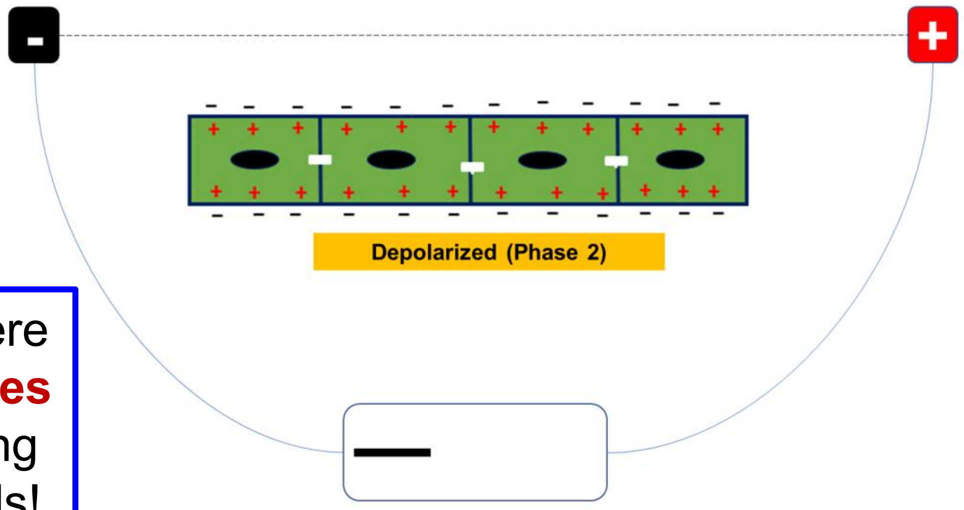
RULES GOVERNING ECG WAVE FORMATION

1. No ECG waves are produced when the atrial or ventricular myocardium is completely repolarized (at rest) or completely depolarized (during plateau phase) state.



TP segment in ECG

Reason: There are **no dipoles** formed during these periods!



ST segment in ECG

These are called ISOELECTRIC lines (or base lines) in ECG

RULES GOVERNING ECG WAVE FORMATION

2. A wave of **depolarization** traveling towards a **positive electrode** of a lead produces a **positive wave** in the ECG tracing of that lead.

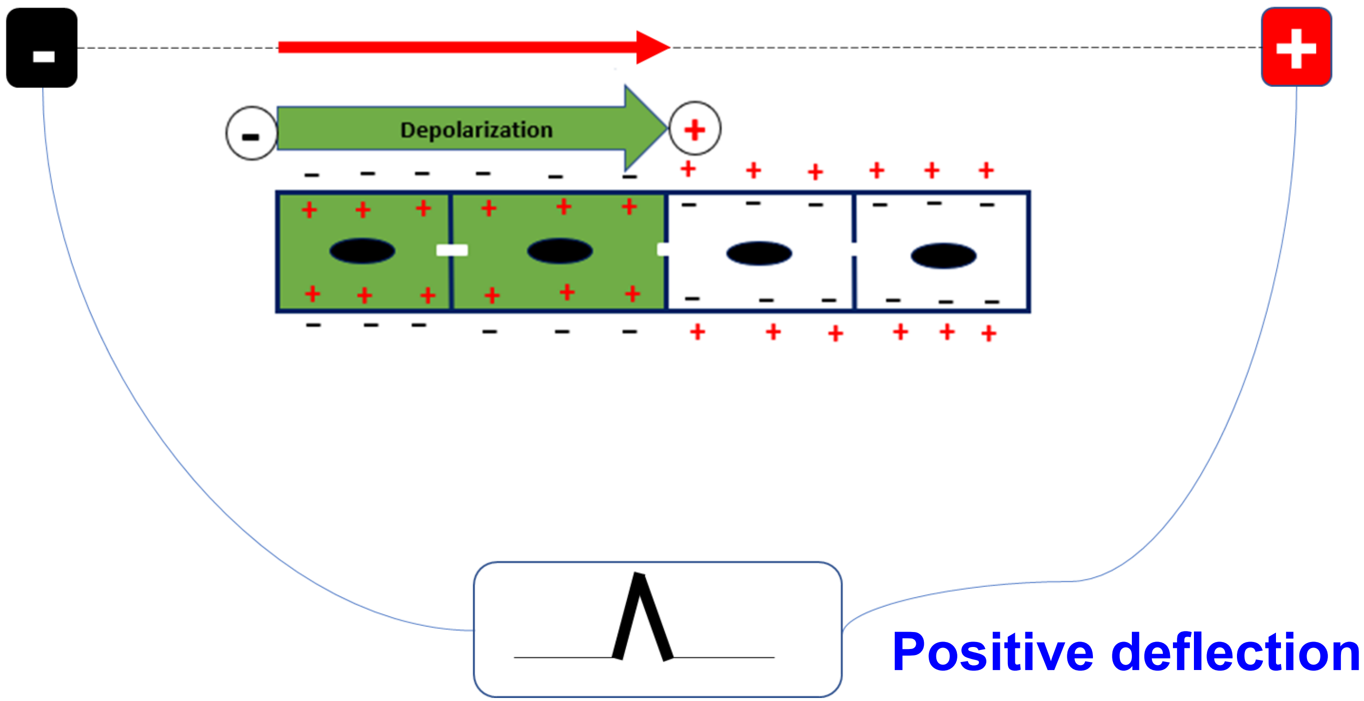
Corollary: A wave of depolarization traveling away from a positive electrode produces a negative deflection.

A wave of repolarization traveling towards a positive electrode results in a negative deflection.

Corollary: A wave of repolarization traveling away from a positive electrode results in a positive deflection.

RULES GOVERNING ECG WAVE FORMATION

Rule # 2: Wave of depolarization moving towards positive electrode



Positive deflection in ECG

Repolarization vector traveling **away** from a Positive Electrode also produces a positive deflection

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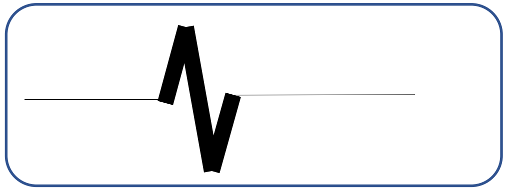
RULES GOVERNING ECG WAVE FORMATION

3. The height of the ECG wave (amplitude of the measured potentials) depends upon the orientation of the Lead axis relative to the **mean electrical vector**:

a) If the Lead axis is **parallel** to the mean electrical vector, a **large** amplitude ECG wave is produced.
b) If the Lead axis is **perpendicular** to the mean electrical vector, an **equiphasic** wave (a combination of positive-negative waves of *equal amplitude*) is produced. Thus, net deflection becomes zero.



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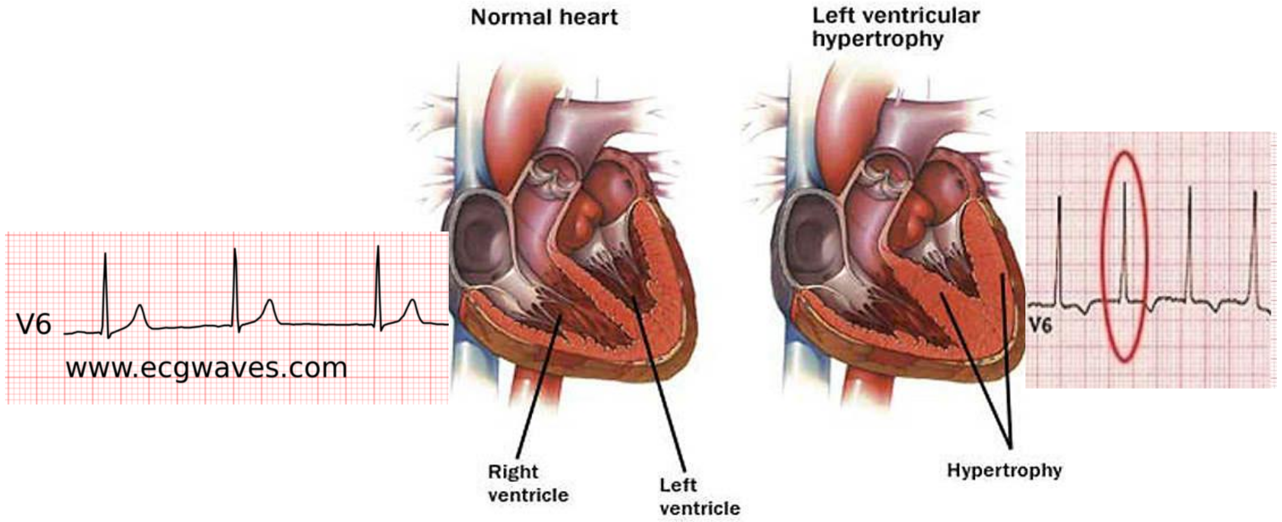
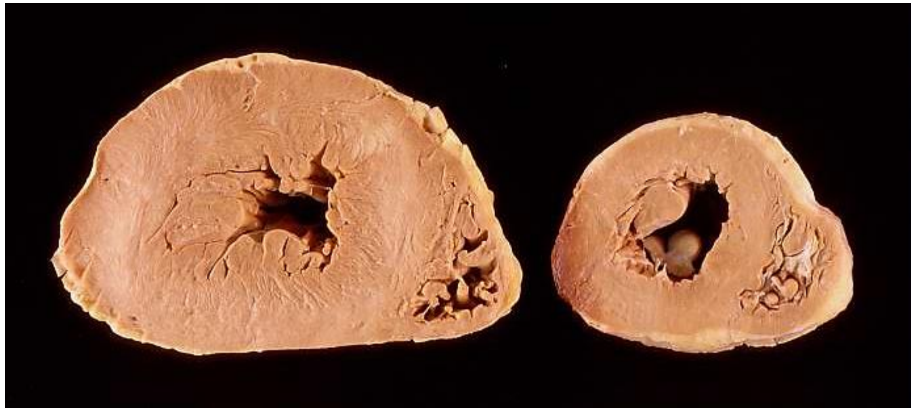
RULES GOVERNING ECG WAVE FORMATION

4. The height of an ECG wave (amplitude), either positive or negative, is directly related to the **mass of the myocardial tissue** undergoing depolarization. If the thickness of a ventricular wall increases, the amplitude increases, as in right or left ventricular hypertrophy.





Hypertrophied LV   Normal LV

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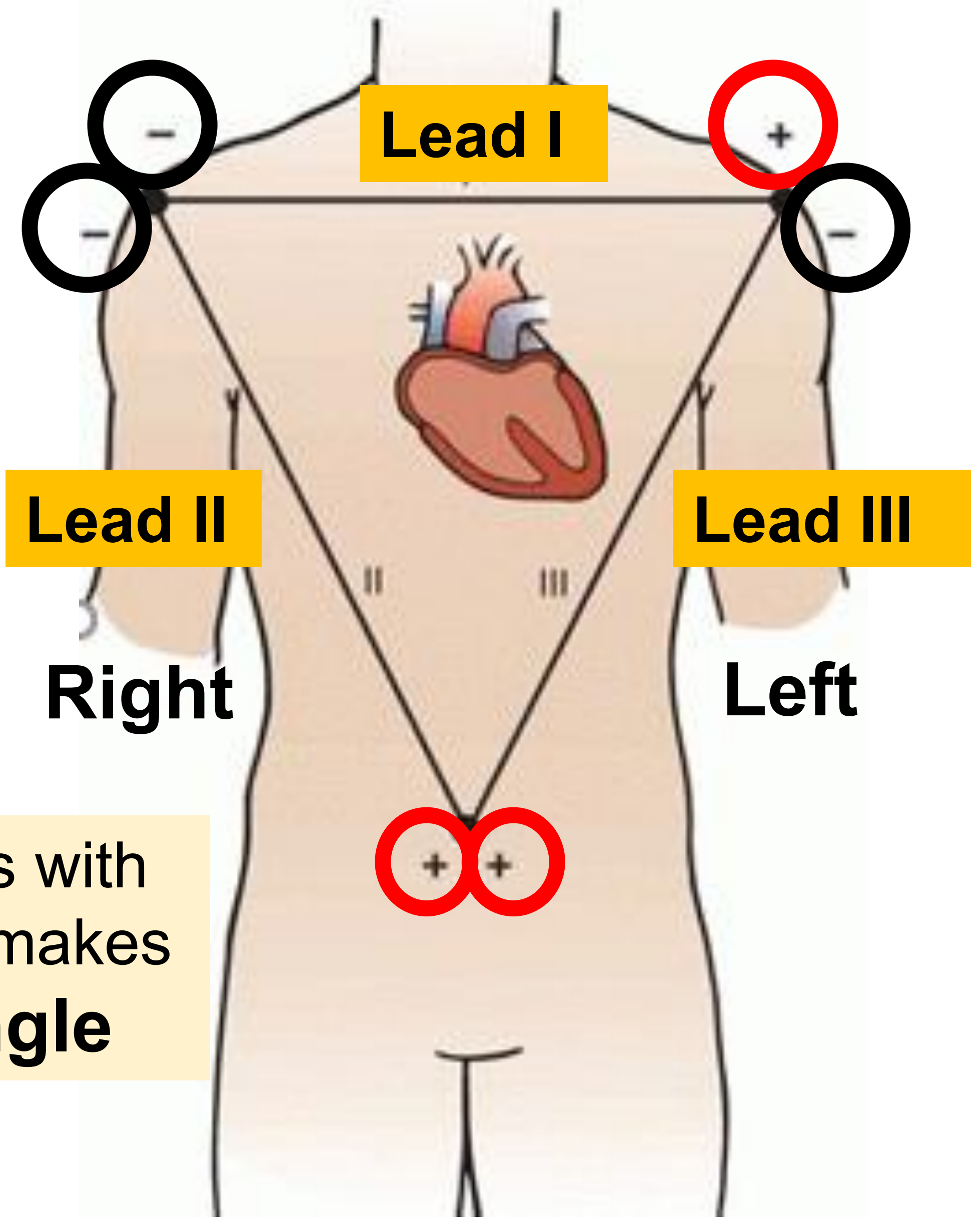
ECG is recorded through Leads

Lead is a combination of electrodes placed at specific sites on the body surface to record ECG



Right arm, Left arm and Left foot are selected for Standard Bipolar Limb Leads, I, II & III

Correct polarity of a lead matters for the origin of an ECG wave

Lead axis refers to an imaginary line between two electrodes

Connecting 3 lead axis with the heart in the center makes **Einthoven's triangle**

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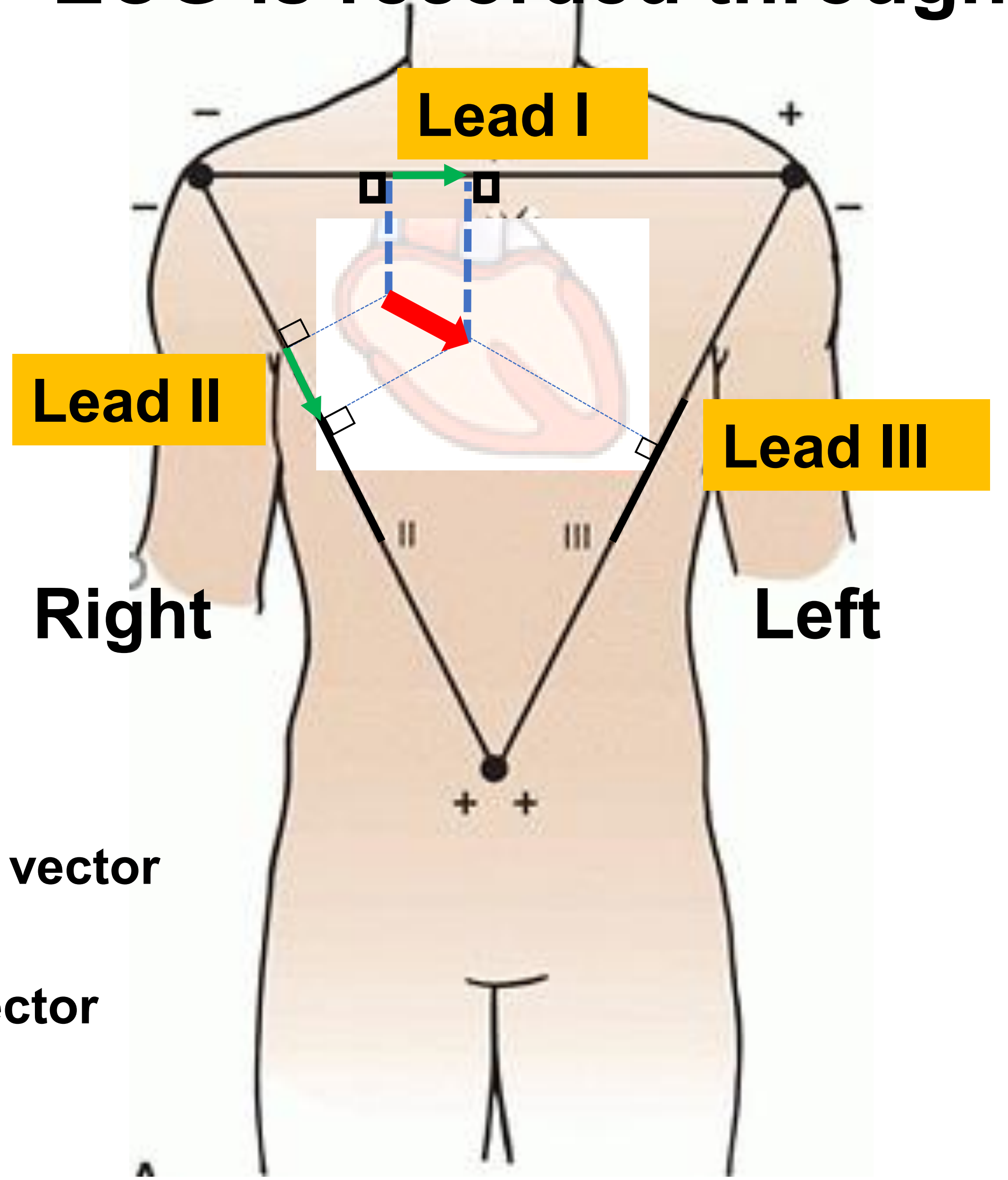
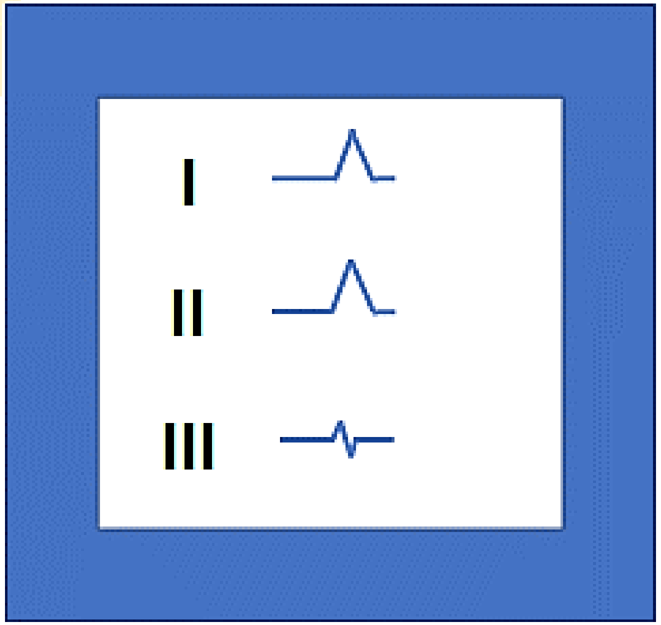
ECG is recorded through Leads



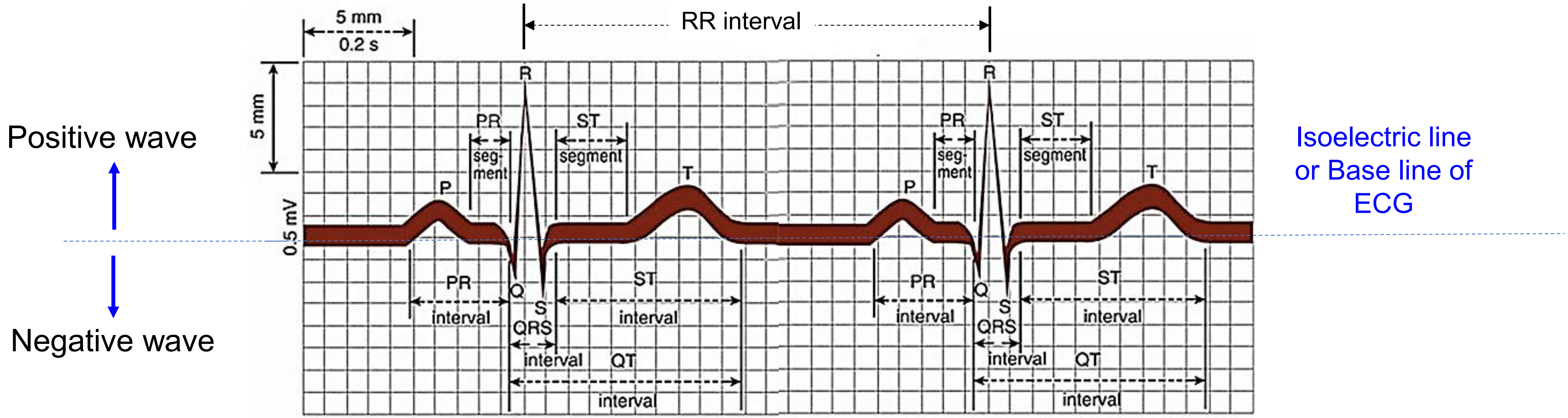
Mean electrical vector is projected on each lead axis and informs about the wave pattern in ECG



Mean electric vector

Projection vector

Normal ECG waves, intervals and segments



Three waves: P, QRS and T

Three segments: PR segment, ST segment and TP segment

Three intervals: PR interval, QRS interval and QT interval

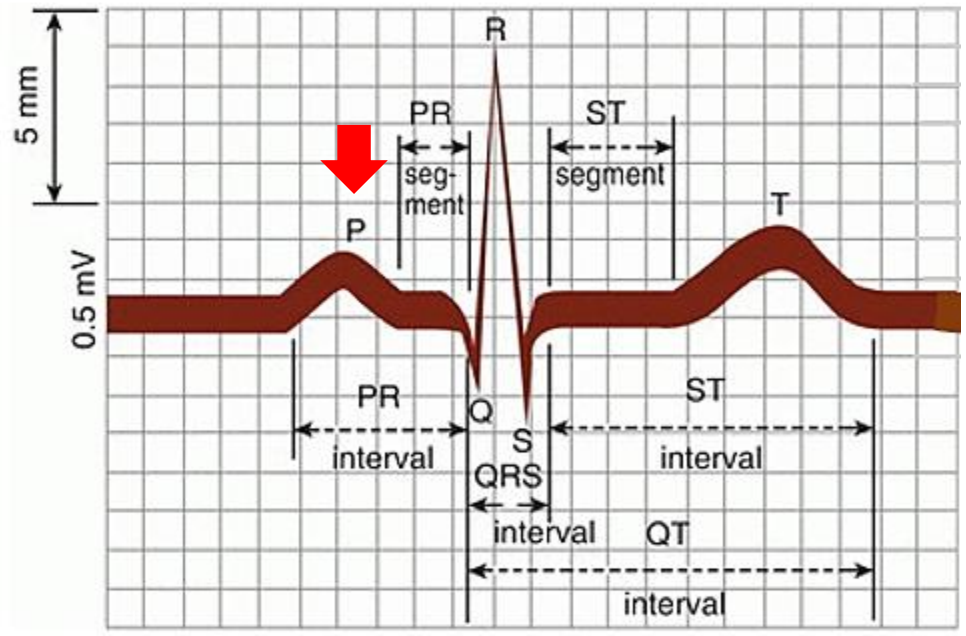
*There is a **fourth interval** called **RR interval** that occurs between two ECG complexes*

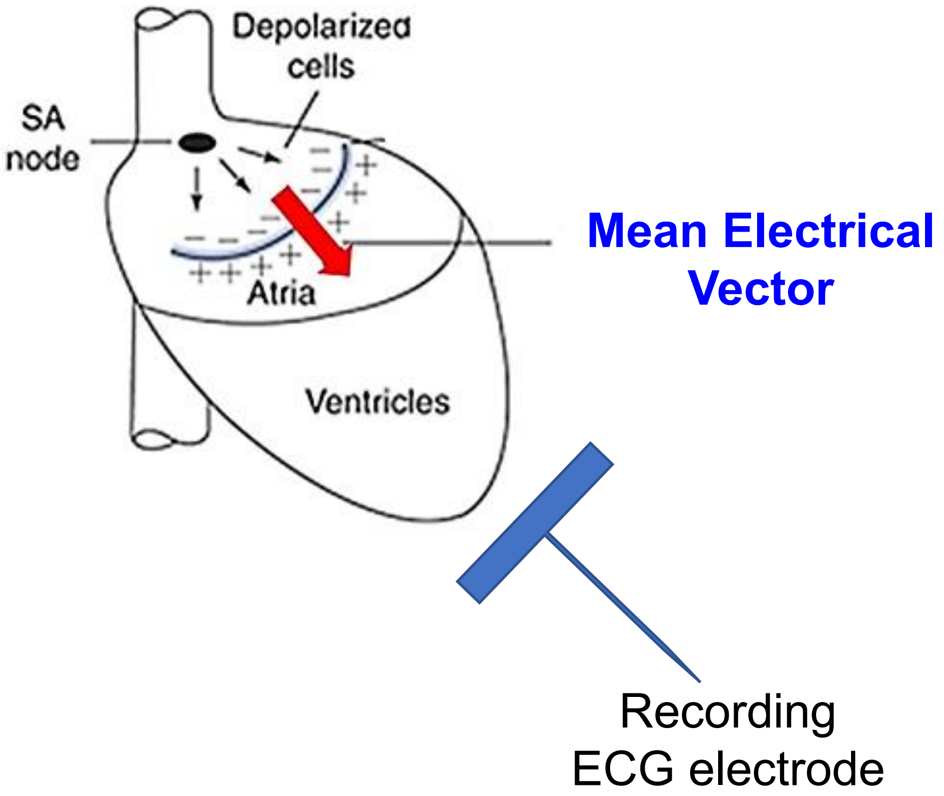
Origin of the ECG waves and segments

ECG wave or ECG segment	Cause or genesis
P wave	During atrial depolarization
PR segment (an isoelectric line)	Occurs when impulse passes through A-V node and Bundle of His
QRS complex (wave)	During ventricular depolarization
ST segment (an isoelectric line)	Occurs when ventricles remain in depolarized state (Phase 2 of the ventricular action potential)
T wave	During ventricular repolarization
TP segment (an isoelectric line)	Occurs when atria and ventricles are at rest with resting membrane potential (Phase 4 of the action potential)

*Atrial repolarization occurs during ventricular depolarization and produces a small electrical wave that is **masked** by a larger ventricular depolarization wave.*

Genesis of ECG waves: P wave origin





P wave: Atrial depolarization wave

Produced when a wave of depolarization spreads from SA node throughout the atria (both right & left) simultaneously.
It is a round wave (compared to sharp QRS waves).

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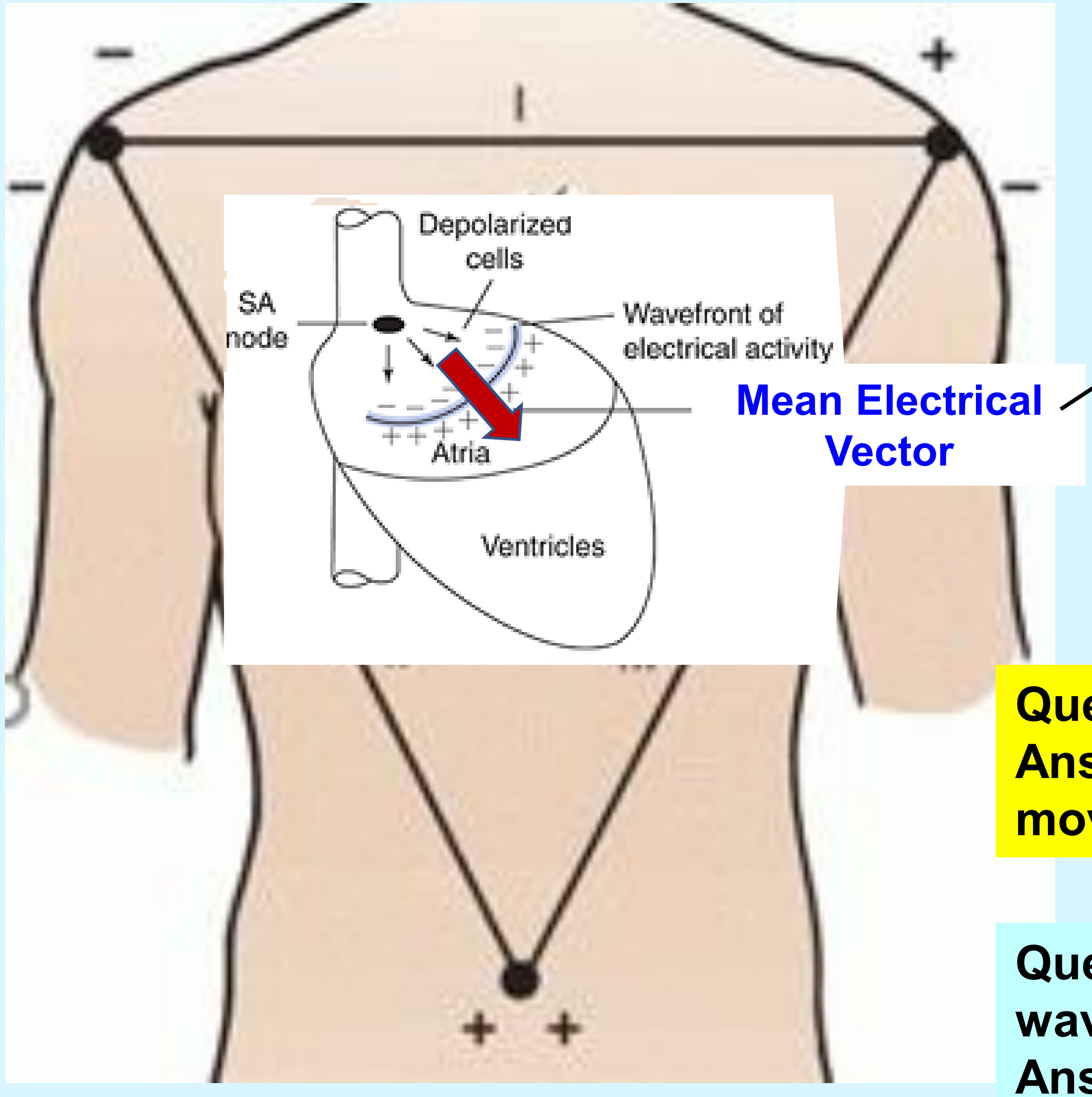
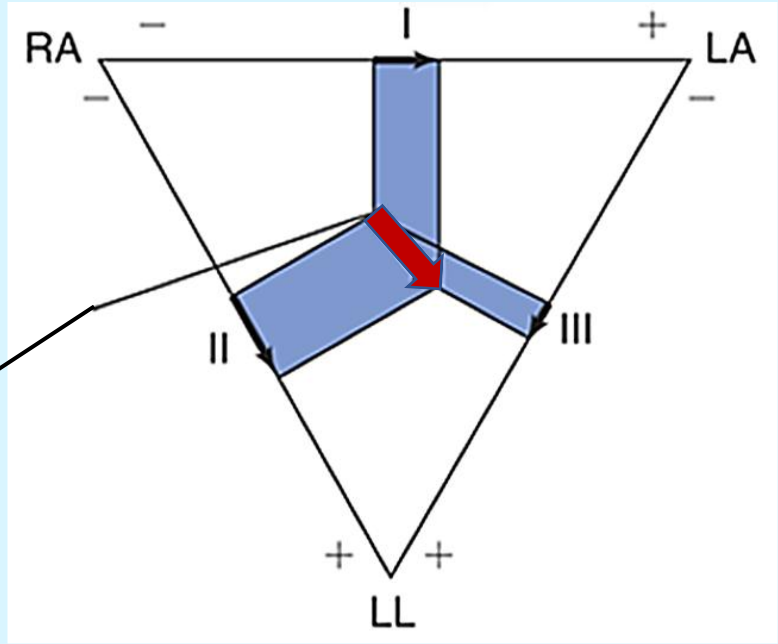
Genesis of ECG waves: P wave origin

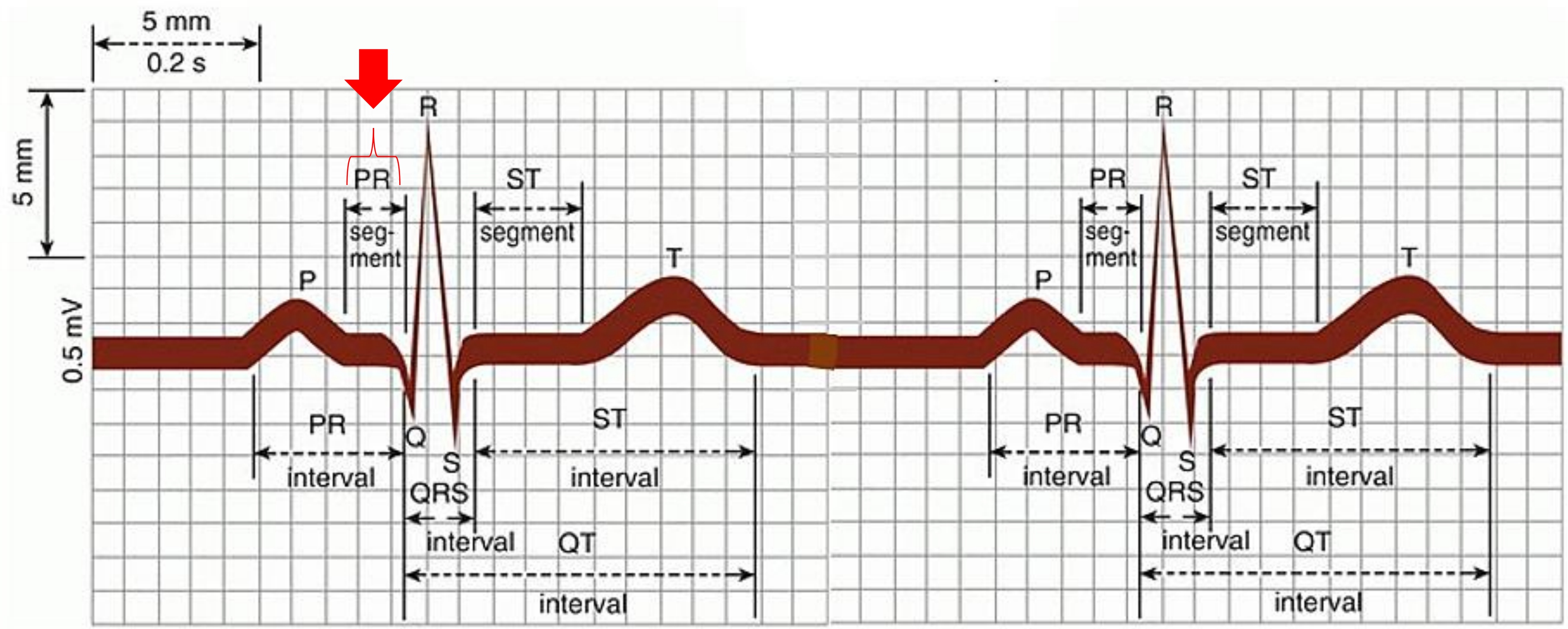




Que: Is the P wave positive in the above 3 Leads?
Ans: Yes, because the wave of depolarization moves towards positive electrodes in all 3 Leads.

Que: Which of the above Leads will have the P-wave with the highest amplitude (height)?
Ans: Lead II, because this lead axis is almost parallel to the mean electrical vector.

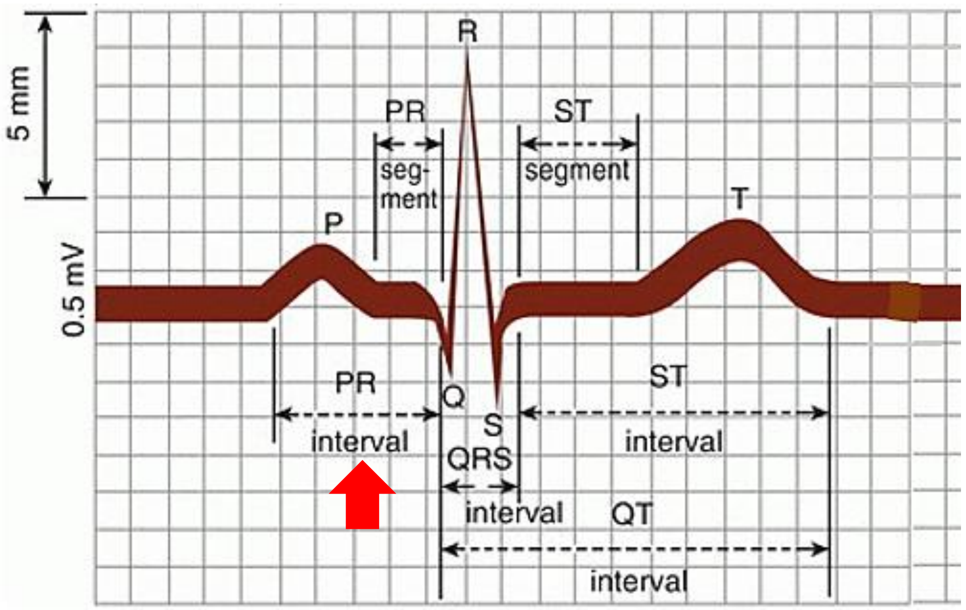
Genesis of ECG waves: PR segment



PR segment:

- Brief **isoelectric** (zero voltage) period after P wave before QRS wave
- Represents the time in which the atrial cells are depolarized, and the impulse is traveling within the AV node.

Genesis of ECG waves: PR interval



PR interval includes time taken for the following events:

- Generation of impulse
- Atrial depolarization
- AV nodal transmission

PR interval :

- Period from the beginning of P wave to the beginning of QRS complex
- Duration: 0.12 – 0.20 sec (3-5 small boxes)
- Represents the time between the onset of atrial depolarization and the onset of ventricular depolarization
- Longer *PR intervals (>0.20 s (>5 small boxes) indicate atrioventricular conduction blocks (useful in diagnosis of* ***Heart blocks****)*

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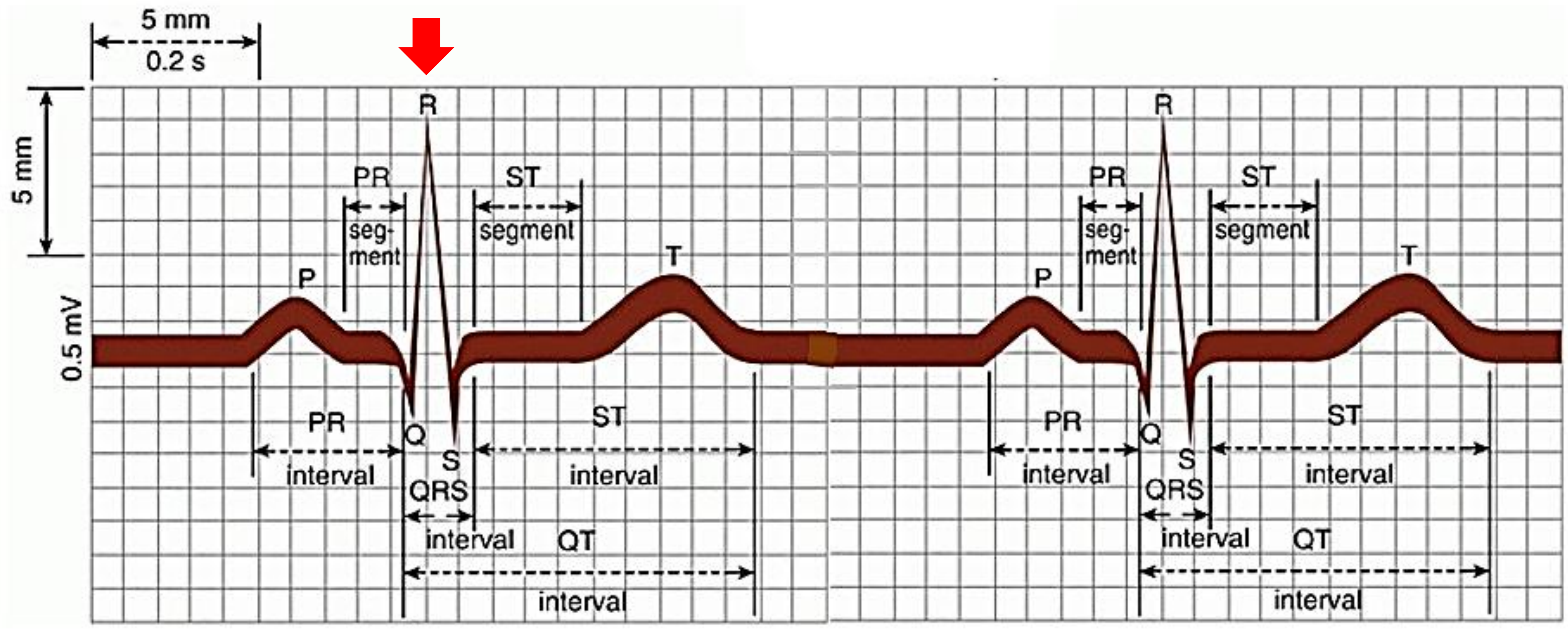
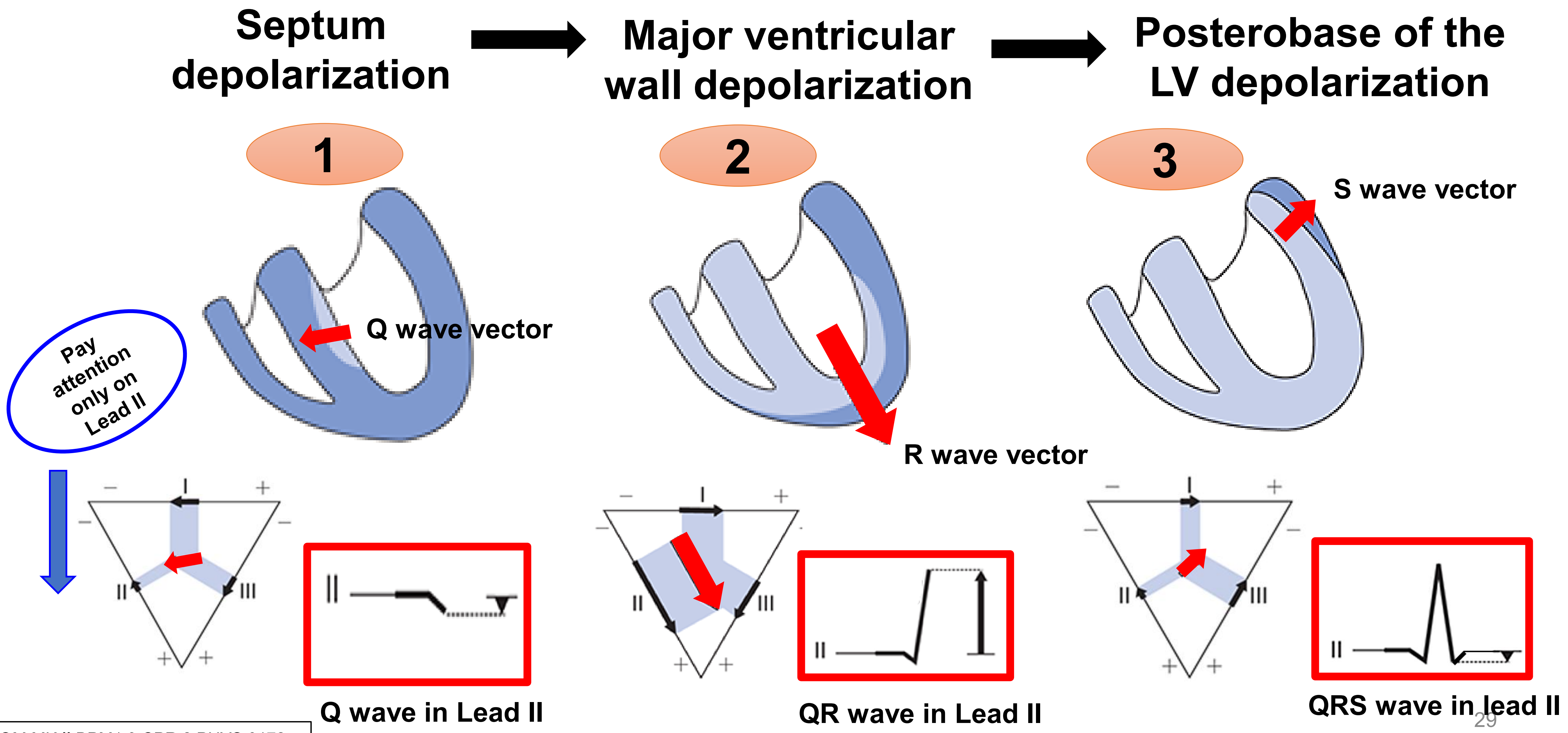
Genesis of ECG waves: QRS complex



QRS complex: Ventricular depolarization wave

- A set of three sharp waves, Q, R and S waves occur together
- Duration (QRS interval): 0.06 – 0.10 sec (Less than 3 small boxes)
- *If QRS complex duration > 0.12 sec (3 small boxes), it is called* ***wide QRS****. Wide QRS occurs in ventricular arrhythmias.*

Genesis of ECG waves: QRS complex origin



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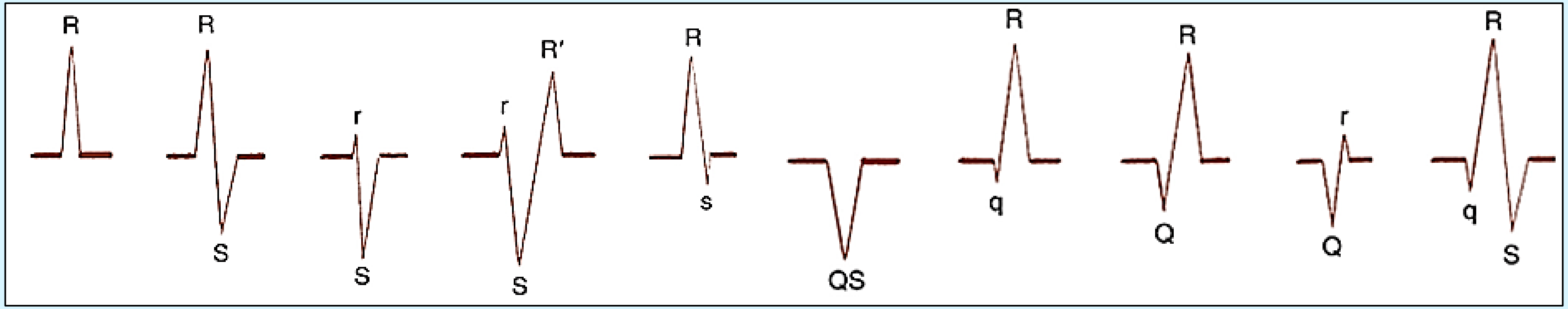
ECG waves: More about labeling QRS waves

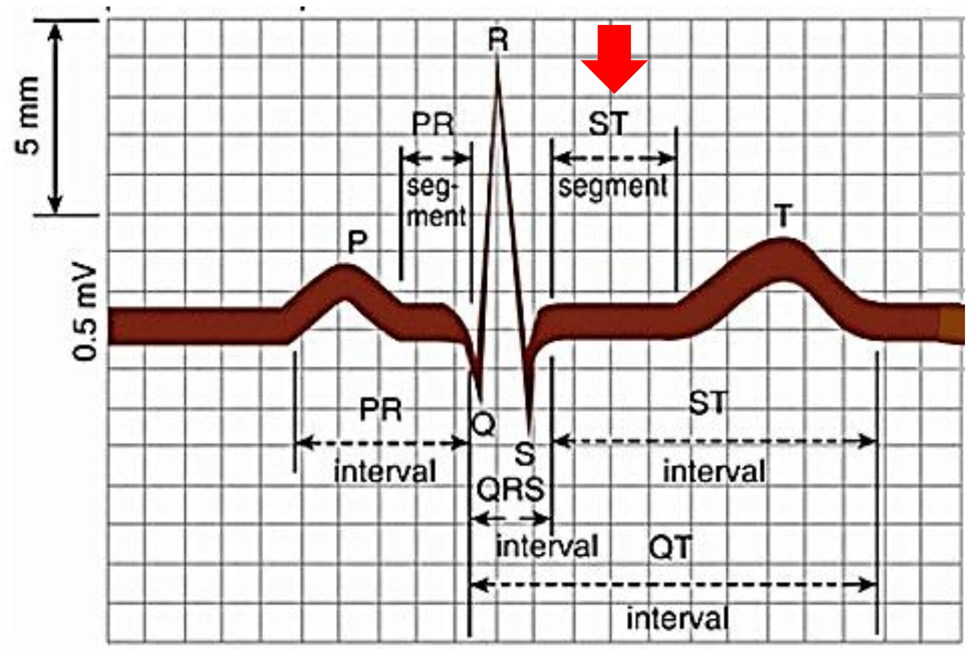
The QRS complex represents the spread of action potentials through the ventricles. This electrical event recorded in every lead DOES NOT contain a Q wave, an R wave, and an S wave. Therefore, the naming goes by the following rule in all leads:

1. When the first initial deflection of the QRS complex is negative, it is called a Q wave.
2. The positive deflection in the QRS complex is called an R wave.
3. A negative deflection following the R wave is called an S wave.

In some leads, there may not be a Q wave present before an R wave. If the entire complex is negative, it is termed a QS wave (not just a Q wave as you might expect). Depending on the amplitude of the wave, the waves may be labeled with upper/lower case (see examples below).



Genesis of ECG waves: ST segment

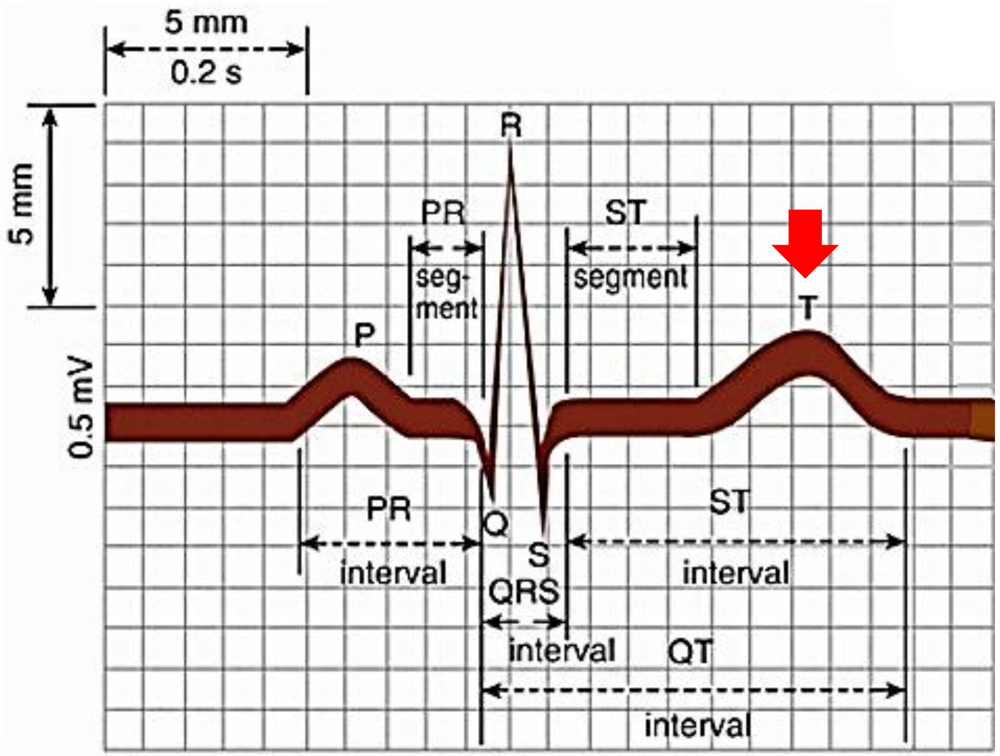


J point (junctional point) is the site where S wave ends, and the ST segment begins. Movement of this point below or above the isoelectric line reflects ST segment depression or elevation, respectively.

ST segment:

- Period following QRS complex when both ventricles are in **depolarized state**
- No dipole formed and hence ST segment is on the isoelectric line
- Corresponds with the plateau phase of the ventricular action potential
- *ST depression or elevation indicates lack of blood supply to the heart. Helps in diagnosis of angina and myocardial infarction.*

Genesis of ECG waves: T wave origin



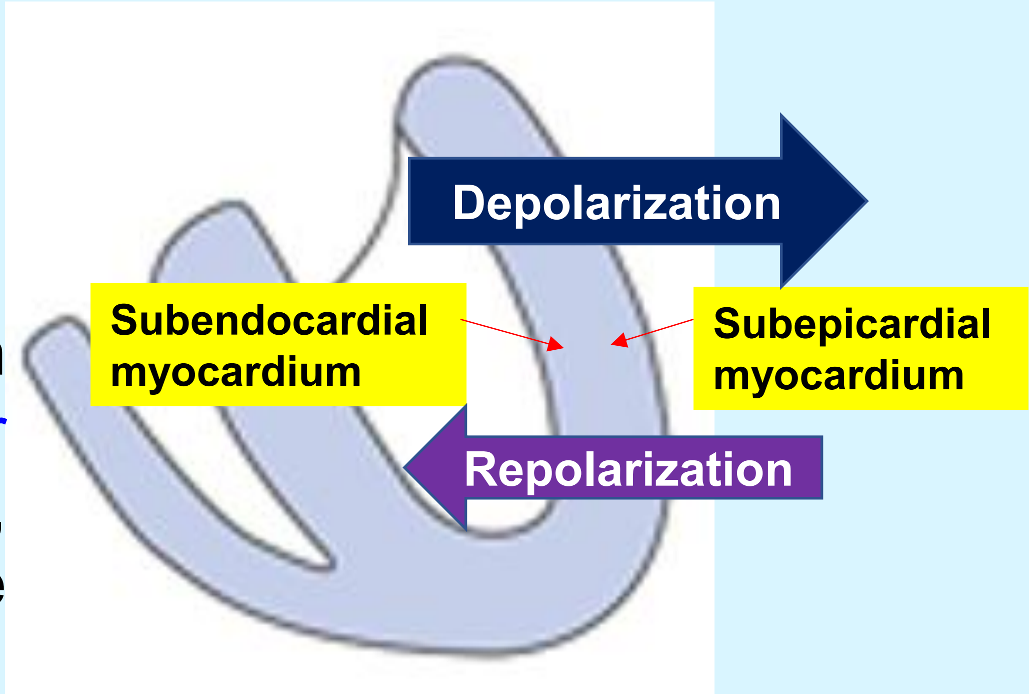
Compare T wave with the QRS waves:
QRS has sharp waves;
T wave is a rounded or gradual wave.

T wave:

- Represents ventricular repolarization (occurs during phase 3 of the ventricular action potential)
- Has a duration longer than that of depolarization and is a gradual wave
- *T-wave inversion indicates lack of blood supply to the myocardium and useful in diagnosis of angina.*

Genesis of ECG waves: Why T wave is also a positive wave

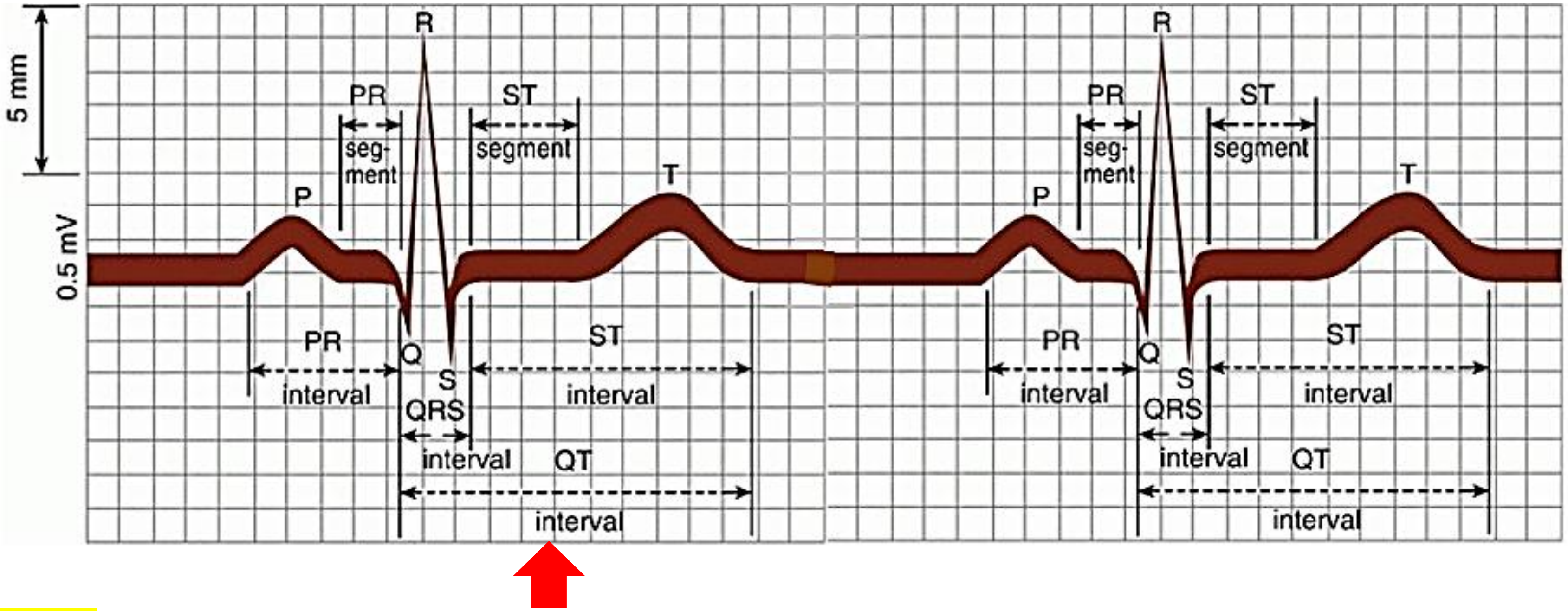
Depolarization of the ventricles occurs first in the subendocardial region and then in the subepicardial region.



Duration of ventricular action potentials in the subendocardial myocardium is longer than in the subepicardial myocardium, therefore, subendocardial myocardium is the first to depolarize and the **last to repolarize**.

This makes the repolarization wave to move away from the recording electrode and therefore, a POSITIVE deflection (T-wave) is produced.

Genesis of ECG waves: QT interval



QT interval :

It is the time taken for ventricular depolarization and ventricular repolarization

Duration: 0.32 – 0.44 sec and varies with heart rate.

In practice, QT interval is expressed as a ***corrected QT (Q-T$_c$) interval***:

$$\textbf{Q-T}_\mathbf{c} = \frac{QT\ interval}{\sqrt{RR\ interval}}$$

Normal Q-T$_c$ interval < 0.44 sec.

> 0.44: prolonged QT interval, high risk of ventricular arrythmias.

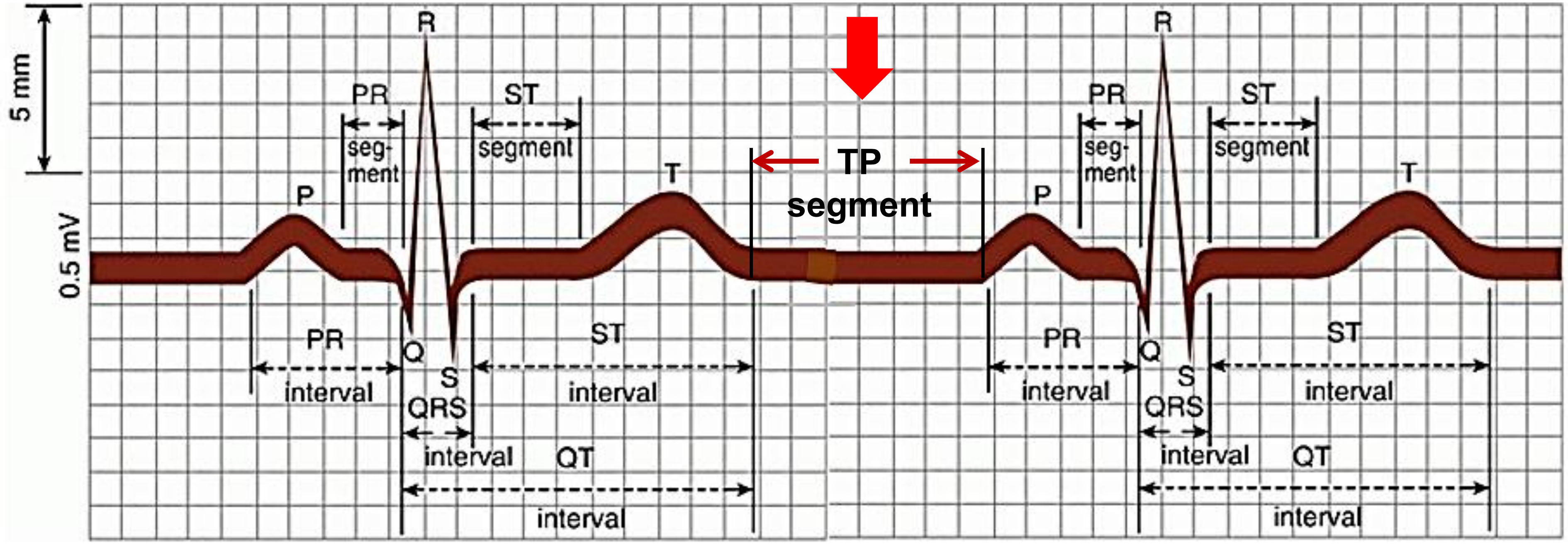
Genesis of ECG waves: Why atrial repolarization wave is NOT seen in an ECG

Atrial repolarization does not appear as a separate deflection in the ECG tracing because:

- It generates a low voltage (slow process) and
- It is **masked** by QRS complex which is a stronger electrical activity that occurs at the same time.

It may be seen when there is complete dissociation between atrial and ventricular electrical activities. If present, it is a negative wave with respect to P wave.

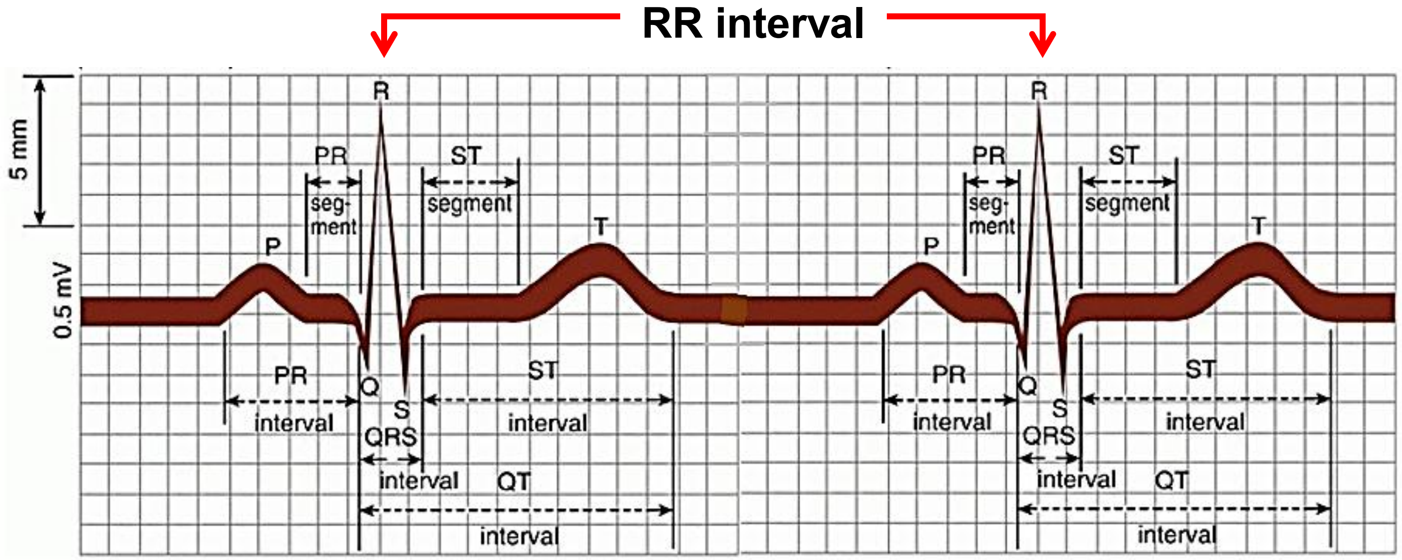
Genesis of ECG waves: TP segment



TP segment:

- Between the end of the T wave and the beginning of the next P wave
- Corresponds to the phase 4 of ventricular Action Potential and represents the electrical resting state
- It is on the isoelectric line (baseline). This is used as the reference baseline to measure the ST-segment depression or elevation.

Additional ECG interval between 2 heart beats: RR interval



Examples

Regular rhythm

Irregular rhythm

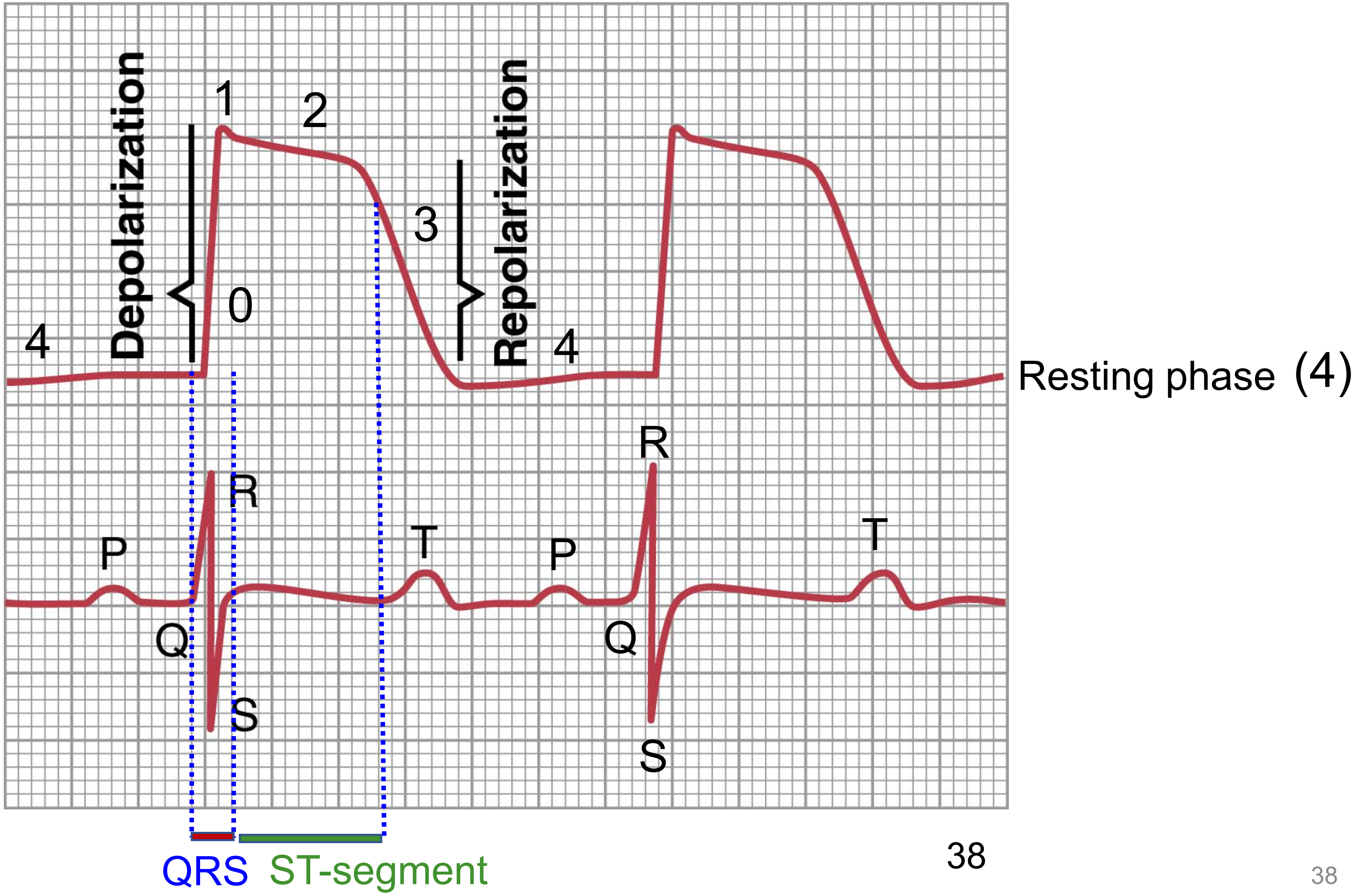
The RR interval:

- It is measured from one R-wave peak of a QRS complex to the corresponding point on the next QRS
- Equivalent to the time take for **one cardiac cycle** (time for one heartbeat).
- Useful to measure heart rate and regularity. Irregular RR intervals indicate irregular heart rhythm (e.g., Atrial fibrillation).

Relationship between ECG and Ventricular Action Potential

Cardiac Action Potential

ECG



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ECG INTERPRETATION (How to read an ECG:steps to follow)

ECG INTERPRETATION (How to read an ECG: steps to follow)

Follow the following 8 steps:

Step 1. Rate: Find the heart rate – Is in normal range? (Normal range is between 60 and 100 bpm)

Step 2. Rhythm: Is the rhythm regular or irregular?

Step 3. Axis: Is there left or right axis deviation?

Step 4. Intervals: What is the duration of the PR interval? What is the QRS interval? What is the QT interval?

Step 5. P wave: What is the height, duration and polarity of the P wave?

Step 6. QRS complex: Is the QRS duration normal or wide?

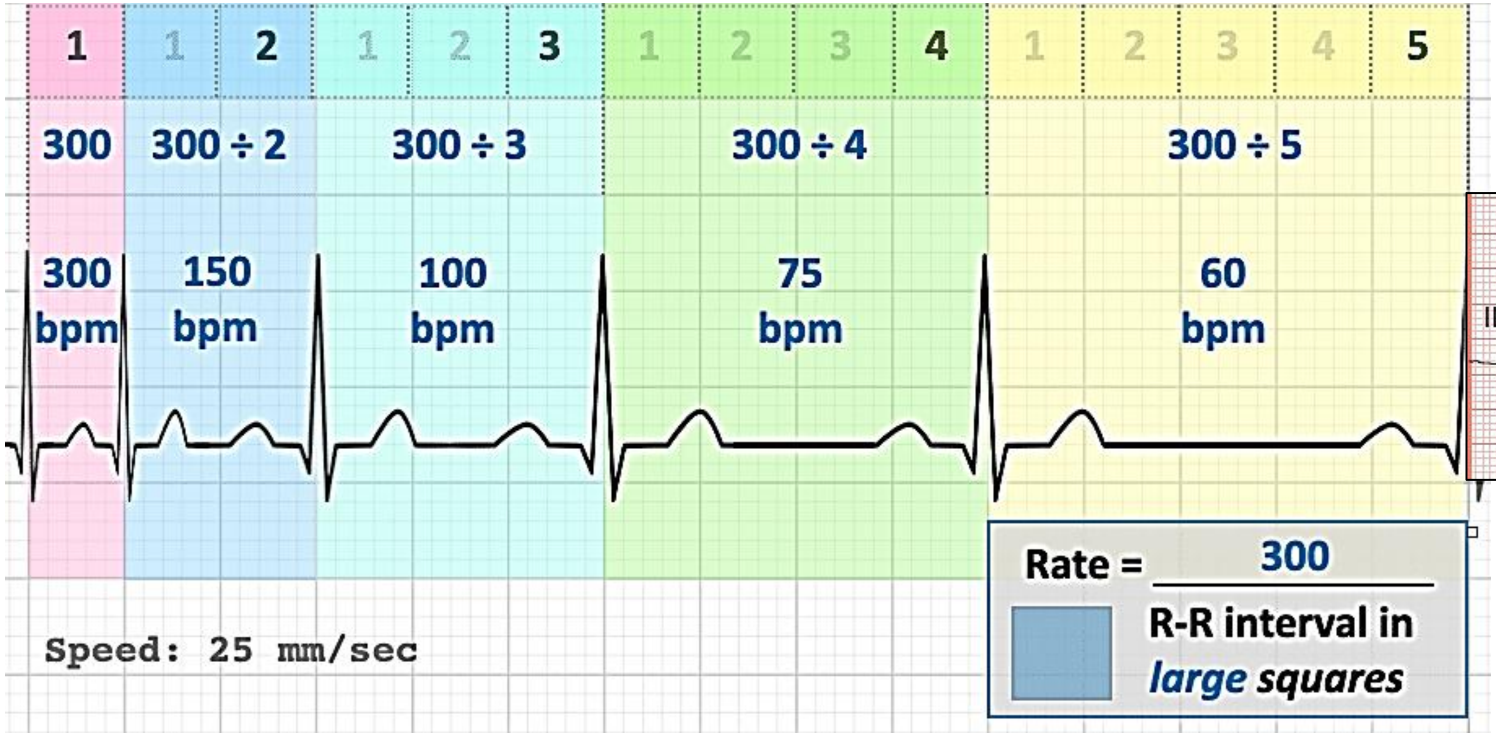
Step 7. ST segment and T wave: Is there ST elevation or depression compared to the TP segment? Are the T waves inverted?

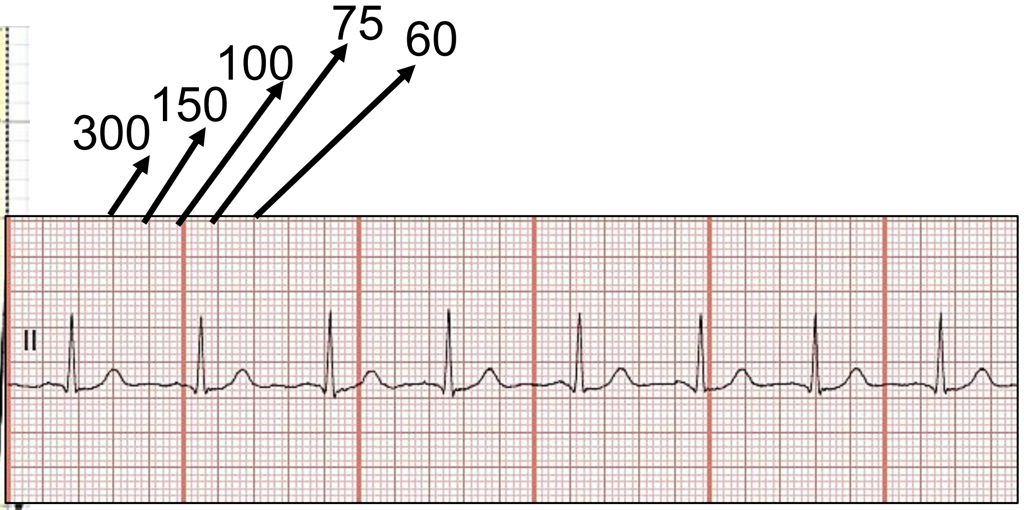
Step 8. Overall interpretation: Give **summary** of the steps, 1-7.

ECG INTERPRETATION: Calculating Heart Rate

Large square method:

- 300 large squares is equal to 1 minute at a normal paper speed of 25 mm/sec.
- We can thus calculate heart rate by dividing 300 by the number of LARGE squares between each R-R interval (space between two consecutive R waves = one beat)





Example:
HR = 300/3.5 large boxes
= 85/min.

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ECG Changes Associated with Myocardial Ischemia

- Ischemia means lack of blood supply to heart muscles.
- Two severities of myocardial ischemia are possible:

i) Ischemia induces chest pain but does NOT cause death of myocytes: Angina.

ii) Ischemia causes death of myocytes: Myocardial Infarction (MI).

- Ischemia creates **current of injury** that shifts ST segment in ECG.

ECG changes in ischemia (Angina)

- ST segment depression
- T wave inversion

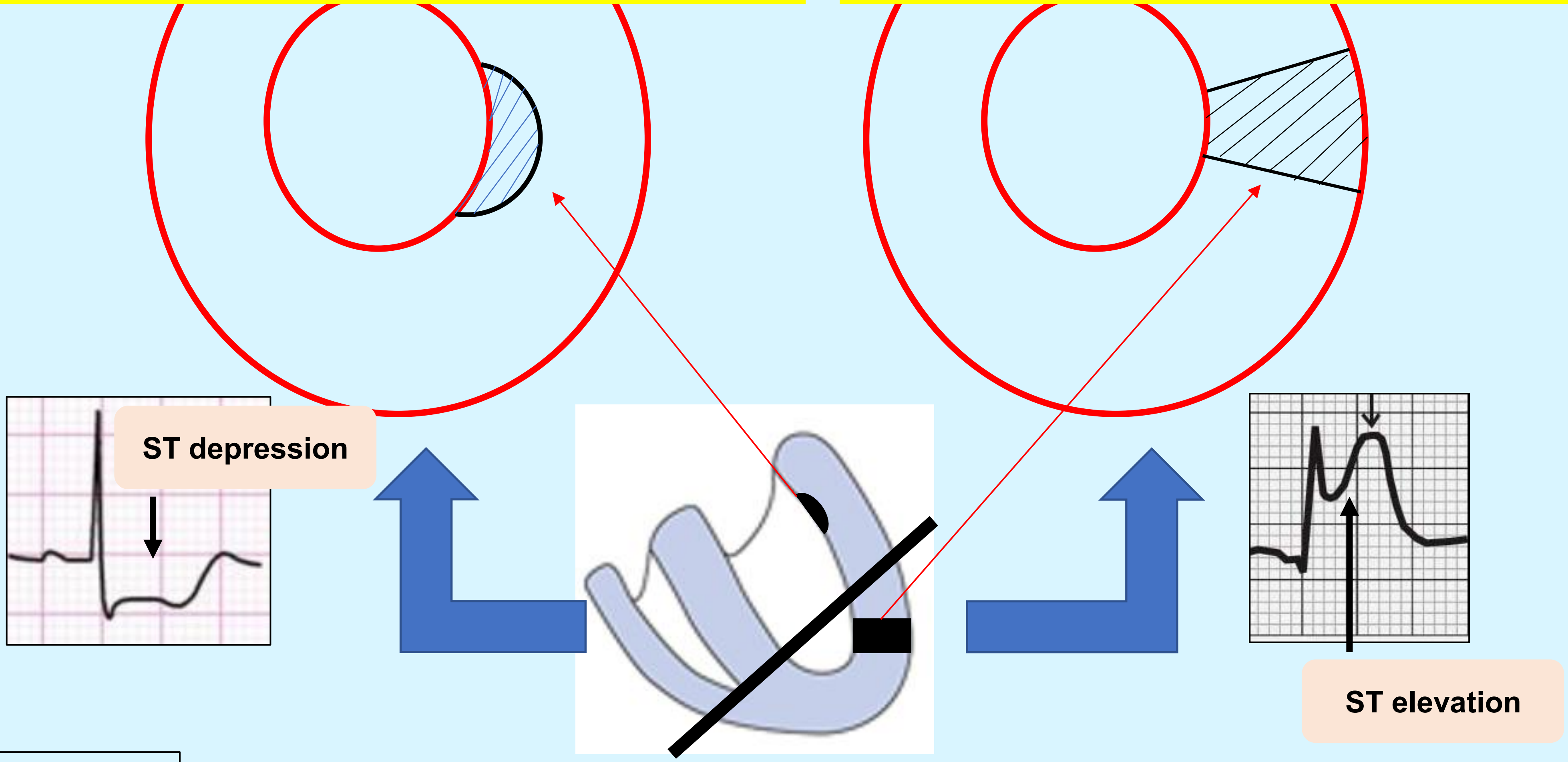
ECG changes in full thickness MI

- ST segment elevation
- Pathological Q waves

(MI with subendocardial injury may not show any ECG changes or show changes similar to angina)

Simple ischemia *(shaded area)* **without infarction produces ST segment depression and a T wave inversion** *(not shown in the fig)*

Full thickness myocardial Infarction *(shaded area)* **produces ST segment elevation**



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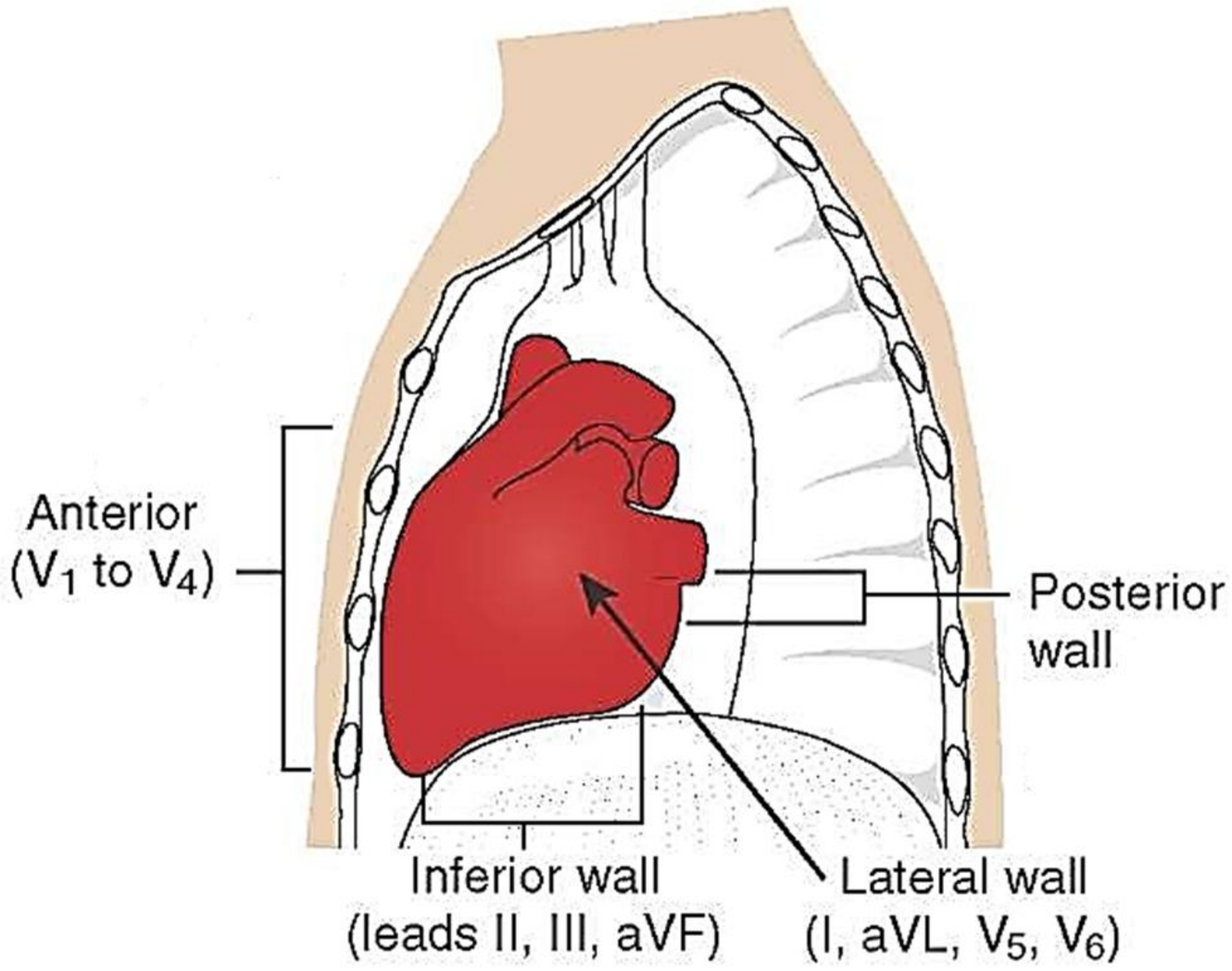
You can locate anatomical site of myocardial injury based on the abnormal ECG in specific Leads



Walls of heart:

- Anterior wall: RV, LV & septum.
- Inferior wall: RV & LV.
- Lateral wall: LV.
- Posterior wall: LV

If you know the anatomical site of lesion, you can predict the coronary vessel that is involved in the injury

Locating anatomical site of myocardial injury based on abnormal ECG in different Leads

This information is required to know for CPRH Small Gp discussion on MI

Location of myocardial injury	Leads that reflect ST seg elevation/depression or abnormal Q wave	Arterial supply involved
Anteroseptal wall	V1-V2	Right coronary and Left Anterior Descending (LAD) arteries
Anteroapical wall	V3-V4	LAD
Anterolateral wall	V5-V6	LAD or Circumflex Artery (LCX)
Lateral wall	I, aVL	LCX
Inferior wall	II, III, aVF	Right Coronary Artery
Posterior wall	V7-V9, Opposite changes in V1-V3 leads	Posterior Descending Artery

Any Questions ?

How to reach me: Either walk into my office or email.

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